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Grade	**0.00** out of 150.00 (**0**%)

Question 1

Incorrect

Mark 0.00 out of 3.00

Descriptive features in psychiatry
Identify the terms used to describe each of the following:

A 70-year-old man lives in a supported accommodation. He goes out for shopping on his own but forgets a few things on his shopping list. When confronted he says that he has already got enough of these items in stock at his flat.

Mitgehen ✖

A 72-year-old male admitted to A&E, when asked what the month was, said 'April' and when asked what the year was, said 'April.'

Choose...

A patient with a moderate intellectual disability is unable to express her emotional problems verbally. When asked if she feels depressed, she says 'yes'. When a picture book with various emotional states is shown, she admits having whatever emotion is shown first in the booklet at different times.

Choose...

Mitgehen
Vorbeigehen
Deja vu
Paraphasia
Suggestibility
Alexithymia
Jamais Vu.
Reflex hallucination
Functional hallucination.
Confabulation
Perseveration

Your answer is incorrect.

Explanation:

Confabulation may be spontaneous where often elements of fantasy are found. It can also occur in an attempt to avoid embarrassment as in this case. Confabulation is the process of describing plausibly false memories for a period for which the patient has amnesia. Two broad patterns emerge: the embarrassed type, in which the patient tries to fill in gaps in memory as a result of awareness of a deficit, and the fantastic type, in which the lacunae are in filled in by details exceeding the need of the memory impairment such as descriptions of wild adventures.

Perseveration is characterised by continuing with a verbal response or action that was initially appropriate after it ceases to be apposite. It is associated with organic brain disease and is occasionally seen in schizophrenia.

This patient is highly suggestible as she answers yes to closed questions and admits to the emotions in any random picture shown to her. Suggestibility is a propensity to be abnormally compliant with suggestions, commands, directions, instructions, or orders from another person.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 105, 115.

Casey P, Kelly B. Fish's clinical pathology: Signs and symptoms in psychiatry, 4th Edition. 2019. p. 67.

Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: A 70-year-old man lives in a supported accommodation. He goes out for shopping on his own but forgets a few things on his shopping list. When confronted he says that he has already got enough of these items in stock at his flat. → Confabulation, A 72-year-old male admitted to A&E, when asked what the month was, said 'April' and when asked what the year was, said 'April.' → Perseveration, A patient with a moderate intellectual disability is unable to express her emotional problems verbally. When asked if she feels depressed, she says 'yes'. When a picture book with various emotional states is shown, she admits having whatever emotion is shown first in the booklet at different times. → Suggestibility

Question 2

Not answered

Marked out of 4.00

MoK Neurosciences CXEMI025

Blood supply to the brain

Match the specific areas of the brain with their blood supply from the list above

1. Hypothalamus ✖

2. Basal ganglia ✖

3. Anterior limb of internal capsule (2 choices) ✖ ✖

Anterior cerebral artery
Anterior communicating artery
Medial cerebral artery
Posterior cerebral artery
Posterior inferior cerebellar artery
Anterior inferior cerebellar artery
Superior cerebellar artery
Anterior inferior cerebellar artery
Internal auditory artery
Subclavian artery

Explanation

Anterior communicating and anterior medial arteries supply the hypothalamus.

The basal ganglia are supplied by the medial cerebral artery.

Anterior cerebral and anterior communicating arteries supply the anterior limb of the internal capsule.

The correct answer is:

Hypothalamus - Anterior communicating artery,

Basal ganglia - Medial cerebral artery,

Anterior limb of internal capsule (2 choices) - Anterior cerebral artery, Anterior communicating artery

Question 3

Not answered

Marked out of 3.00

Personality disorders - diagnosis

Choose one option each from above list:

A 35-year-old man refuses to provide answers to standard questions during an initial clerking and threatens to stop the interview if recording his telephone number is insisted. It seems that he has taken many hospitals to court for suspicions about how his personal data has been handled.

A 33-year-old night security at a local hospital prefers to be alone whenever possible. He has no friends, and he does not socialise. He does not keep update with current affairs and has no sexual interests. He spends most time day dreaming.

A 21-year-old lady spends excessive time seeking attention by making herself attractive. She wears the 'fashion of the week' all the time and show shallow but labile affect. She comes across as being very impressionistic.

Histronic personality disorder
Obsessive-compulsove personality disorder
Narcissistic personality disorder
Antisocial personality disorder
Paranoid personality disorder
Schizoid personality disorder
Dependent personality disorder
Boderline personality disorder
Schizotypal personality disorder
Avoidant personality disorder

Choose...

Your answer is incorrect.

Explanation:

Clinical features of patients with paranoid personality disorder include being suspicious that others are exploiting, harming, or deceiving them; reading hidden demeaning or threatening meanings into benign remarks/events; holding long-standing grudges; believing that others are not loyal or trustworthy; being reluctant to confide in others; feeling that others are attacking their character or reputation; having recurrent suspicions of infidelity.

Clinical features of patients with schizoid personality disorder include not enjoying relationships; preferring to do things alone; having no interest in sexual relations; experiencing no enjoyment of activities; having few, if any, friends or acquaintances; not caring what others think of them; being emotionally constricted.

Clinical features of patients with histrionic personality disorder include being overly dramatic and theatrical; considering relationships more intimate than they are; using physical appearance to draw attention; speaking in a way that is excessively impressionistic and lacking in detail; having shallow and shifting emotions; being inappropriately sexually provocative; feeling uncomfortable when not the centre of attention; being suggestible and easily influenced.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 562-565.

The correct answer is: A 35-year-old man refuses to provide answers to standard questions during an initial clerking and threatens to stop the interview if recording his telephone number is insisted. It seems that he has taken many hospitals to court for suspicions about how his personal data has been handled. → Paranoid personality disorder, A 33-year-old night security at a local hospital prefers to be alone whenever possible. He has no friends, and he does not socialise. He does not keep update with current affairs and has no sexual interests. He spends most time day dreaming. → Schizoid personality disorder, A 21-year-old lady spends excessive time seeking attention by making herself attractive. She wears the 'fashion of the week' all the time and show shallow but labile affect. She comes across as being very impressionistic. → Histrionic personality disorder

Question 4

Not answered

Marked out of 4.00

Neuroimaging tests in clinical practice

Identify the most likely neuroimaging findings for the following neuropsychiatric conditions

A 22-year-old gentleman with persecutory delusions and 3rd person auditory hallucinations. In addition to reduced frontotemporal and insular grey matter, what other feature can be seen in structural imaging?

Choose...

A 69-year-old woman with recurrent episodes of low mood, anhedonia and poor sleep since the age of 60. What feature can be seen in structural imaging?

Choose...

A 77-year-old man with history of short-term memory loss and poor concentration. In addition to ventricular enlargement, what other feature can be seen in structural imaging?

Choose...

A 33-year-old woman with several waxing and waning neurological deficits involving ocular movements and sensorimotor system but intact cognitive abilities.

Choose...

- White matter hyperintensities
- Progressive increase in brain volume in first few years of diagnosis
- Medial temporal atropy
- Ventricular enlargement
- Loss of white matter in the dorsolateral prefrontal cortex
- Lower prefrontal metabolism
- White matter hypointensities
- Increased frontal lobe volumes

Your answer is incorrect.

Explanation:

Case 1: The diagnosis is schizophrenia. Neuroimaging findings in schizophrenia include ventricular enlargement and loss of grey matter - especially insula, anterior cingulate cortex, medial temporal lobe and dorsolateral pre-frontal cortex with some progressive loss of brain volume in the first few years of diagnosis. fMRI shows poor dorsolateral pre-frontal cortex activation in executive tasks, while decreased N-acetyl aspartate in prefrontal cortex is seen in spectroscopy.

Case 2: The diagnosis is depressive disorder (geriatric depression). Neuroimaging findings include periventricular and deep WM hyperintensities, subcortical thalamic and striatal hyperintensities, decreased frontal and basal ganglia volumes, and decreased metabolism in prefrontal cortex.

Case 3: The diagnosis is Alzheimer's dementia. Neuroimaging findings in Alzheimer's disease include ventricular enlargement, reduced medial temporal lobe volume - especially hippocampus and a reduction in parieto-temporal fMRI activation and SPECT blood flow.

Case 4: The diagnosis is Multiple Sclerosis. Neuroimaging findings in this disease include features of demyelination, especially white matter hypointensities affecting corpus callosum. If fluid-attenuated inversion recovery (FLAIR) MRI is used, periventricular high-signal intensity lesions can be observed.

The correct answer is: A 22-year-old gentleman with persecutory delusions and 3rd person auditory hallucinations. In addition to reduced frontotemporal and insular grey matter, what other feature can be seen in structural imaging? → Ventricular enlargement, A 69-year-old woman with recurrent episodes of low mood, anhedonia and poor sleep since the age of 60. What feature can be seen in structural imaging? → White matter hyperintensities, A 77-year-old man with history of short-term memory loss and poor concentration. In addition to ventricular enlargement, what other feature can be seen in structural imaging? → Medial temporal atrophy, A 33-year-old woman with several waxing and waning neurological deficits involving ocular movements and sensorimotor system but intact cognitive abilities. → White matter hypointensities

Question 5

Not answered

Marked out of 4.00

Neuroanatomical lesions in CNS disorders

For each of the condition below choose the most specific option/s from given list

Korsakoff psychosis	Choose...
Gerstmann syndrome	Choose...
Prosopagnosia	Choose...
Alzheimer disease	Choose...

Occipitotemporal junction
Corpus callosum
Inferior frontal gyrus
Dorsolateral prefrontal cortex
Third ventricle
Dorsomedial thalamus
Supramarginal gyrus
Hippocampus
Hypothalamus

Your answer is incorrect.

Explanation:

About 80 per cent of alcoholic patients recovering from Wernicke encephalopathy develop Korsakoff amnesic syndrome. Diagnosis of Korsakoff syndrome correlates best with the presence of lesions in the dorsomedial thalamus. But lesions in the mammillary bodies, the mammillothalamic tract and the anterior thalamus may be more relevant to memory dysfunction than lesions in the medial dorsal nucleus of the thalamus (Kopelamn et al., 2009).

Gerstmann syndrome is characterized by four primary symptoms: Dysgraphia/agraphia, Dyscalculia/acalculia, Finger agnosia and Left-right disorientation. This is often associated with brain lesions in the dominant (usually left) side of the angular and supramarginal gyri (parietal lobe).

Prosopagnosics cannot recognise familiar faces. Face processing is a bilateral function; more vital areas may be present on the right hemisphere. Acquired prosopagnosia is usually associated with bilateral or right-sided lesions of the occipital - temporal junction (fusiform gyrus).

In Alzheimer disease, the main areas affected are the hippocampus and medial temporal lobe. Atrophic changes are mainly observed in these two areas in even in very early stages of Alzheimer disease.

Barton, JJ., (2008). Prosopagnosia associated with a left occipitotemporal lesion. Excerpt retrieved from http://www.ncbi.nlm.nih.gov/pubmed/18374372
Kopeman et al., (2009). The Korsakoff Syndrome: Clinical Aspects, Psychology and Treatment. Excerpt retrieved from http://alcalc.oxfordjournals.org/content/44/2/148

The correct answer is: Korsakoff psychosis → Dorsomedial thalamus, Gerstmann syndrome → Supramarginal gyrus, Prosopagnosia → Occipitotemporal junction, Alzheimer disease → Hippocampus

Question 6

Not answered

Marked out of 1.00

Which of the following antipsychotics is the fastest to dissociate from the dopamine D2 receptor?

Select one:

- Aripiprazole
- Xanomeline
- Clozapine
- Paliperidone
- Risperidone

Your answer is incorrect.

Explanation:

Clozapine is an antagonist of 5-HT2A, D1, D3, and D4. It also has potent antagonism of muscarinic M1, H1, and alpha-1 adrenergic receptors. It has relatively low potency as a D2 receptor antagonist. At therapeutic doses, clozapine occupies fewer D2 receptors than the other drugs that treat psychosis. Clinically effective doses of clozapine occupy only 40-50% of striatal D2 receptors. This is probably why clozapine has low propensity to cause EPS. It has also been postulated that clozapine binds more loosely to the D2 receptor, and because of this 'fast dissociation', more normal dopamine neurotransmission is possible.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 613.

Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 225.

The correct answer is: Clozapine

Question 7

Not answered

Marked out of 1.00

Which of the following street drugs has local anaesthetic properties?

Select one:

- Ketamine
- Cannabis
- LSD
- Cocaine
- Heroin

Your answer is incorrect.

Explanation:

Cocaine has membrane stabilising, and thus local anaesthetic, effects. It is a local anaesthetic which Freud himself exploited to help dull the pain of his tongue cancer.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 545.

The correct answer is: Cocaine

Question 8

Not answered

Marked out of 1.00

A 41-year-old woman is taking disulfiram to aid abstinence from alcohol. She present to A and E with sickness and feeling unwell. On examination she appears to have facial flushing with low BP but high pulse rate. She denies having drunk alcohol. Which of the following might explain her symptoms?

Select one:

- Mature cheese
- 2 mugs of black coffee
- Cough syrup
- Banana yoghurt
- Tramadol tablets

Your answer is incorrect.

Explanation:

Many cough and common cold aliments contain a significant amount of alcohol sufficient to precipitate an alcohol-disulfiram reaction. If alcohol is consumed by a patient taking disulfiram, toxic levels of acetaldehyde build up, causing unpleasant side effects. The patient should be advised not to consume any food or beverages containing alcohol, or to use any alcohol-containing preparations (e.g. cough syrup).

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. pp. 239-240.

The correct answer is: Cough syrup

Question 9

Not answered

Marked out of 1.00

The antiemetic property of chlorpromazine is mainly due to

Select one:

- D2 blockade
- 5HT2A blockade
- Antiadrenergic effect
- Anticholinergic effect
- 5HT3 blockade

Your answer is incorrect.

D2 blockade due to chlorpromazine in chemoreceptor trigger zone explains its antiemetic property.

The correct answer is: D2 blockade

Question 10

Not answered

Marked out of 1.00

Sally is taking amisulpride and is complaining of amenorrhea, galactorrhoea and loss of sexual desire. This effect is caused by

Select one:

- 5HT1A antagonism
- D1 blockade
- D2 blockade
- D3 blockade
- 5HT2A antagonism

Your answer is incorrect.

Explanation:

When D2 antagonists are administered, D2 receptors in the tuberoinfundibular dopamine pathway are blocked. This causes plasma prolactin concentrations to rise, a condition called hyperprolactinaemia. This can be associated with galactorrhea (i.e. breast secretions), sexual dysfunction, gynaecomastia (enlargement of the breasts) in men, and amenorrhea (i.e. irregular or lack of menstrual periods) in women.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 164-165.

The correct answer is: D2 blockade

Question 11

Not answered

Marked out of 1.00

Which of the following is a known psychiatric effect of the drug clonidine?

Select one:

- Psychosis
- Mania
- Anorexia
- Depression
- Dementia

Your answer is incorrect.

Explanation:

The main adverse effects of clonidine are sedation, postural hypotension, and depression. Other drugs that can cause depression include propranolol, digoxin, frusemide, reserpine, cimetidine, methyldopa, bromocriptine, and phenothiazines.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 588.

The correct answer is: Depression

Question 12

Not answered

Marked out of 1.00

The term "intrinsic activity" of a drug-receptor complex refers to

Select one:

- The efficiency of a drug in reaching the appropriate receptors
- Potency of the drug to cause a functional response
- Affinity of the drug to the receptor
- Efficacy of the resulting functional response
- The receptor action in the absence of the drug molecule

Your answer is incorrect.

Intrinsic activity refers to the capability of a drug-receptor complex to produce the highest functional response (i.e. efficacy).

The correct answer is: Efficacy of the resulting functional response

Question 13

Not answered

Marked out of 1.00

Gamma hydroxybutyrate (GHB) acts on which of the following receptors?

Select one:

- Dopamine
- Opioid
- GABA-B
- Serotonin
- Noradrenaline

Your answer is incorrect.

Explanation:

Sodium oxybate is also known as gamma-hydroxybutyrate (GHB), and acts as a full agonist at GHB receptors and a partial agonist at GABA-B receptors. GHB is a natural product in the brain and formed by the neurotransmitter GABA. It has been labeled a 'date rape' drug as it can be used with alcohol for this purpose, knocking someone out and causing amnesia for the time while involuntarily intoxicated. Because it profoundly increases slow-wave sleep and the growth hormone surge that accompanies slow wave sleep, it was also used (abused) by athletes as a performance-enhancing drug, especially in the 1980s when it was sold over the counter in health food stores. GHB is used in some European countries as a treatment for alcoholism.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. pp. 446-448.

The correct answer is: GABA-B

Question 14

Not answered

Marked out of 1.00

A patient who has had a renal transplant is currently on cyclosporine for immunosuppression. He appears to be clinically depressed. The renal physician says it has been difficult titrating the cyclosporine levels. Which of the following drugs must be avoided as the first line intervention?

Select one:

- amitriptyline
- fluoxetine
- venlafaxine
- citalopram
- phenelzine

Your answer is incorrect.

Fluoxetine, fluvoxamine and nefazodone are potent CYP3A4 inhibitors and can significantly affect cyclosporine levels in patients.

The correct answer is: fluoxetine

Question 15

Not answered

Marked out of 1.00

A 60-year-old gentleman has a diagnosis of bipolar affective disorder and is on Lithium carbonate. He has recently developed systemic hypertension. Which one among the following is the most appropriate drug to treat his hypertension?

Select one:

- Bendroflumethiazide
- Amiloride
- Captopril
- Furosemide
- Chlorthalidone

Your answer is incorrect.

Explanation:

ACE inhibitors (e.g. captopril) can reduce thirst (which can lead to mild dehydration) and increase renal sodium loss (leading to increased sodium re-absorption by the kidney), resulting in an increase in lithium plasma levels. The magnitude of this effect is variable, from no increase to a four-fold increase. Diuretics can reduce the renal clearance of lithium, the magnitude of this effect being greater with thiazide than loop diuretics. Lithium levels usually rise within 10 days of a thiazide diuretic (e.g. bendroflumethiazide, chlortalidone, inadapamide, xipamide) being prescribed. Although there are case reports of lithium toxicity induced by loop diuretics, many patients receive loop diuretics (e.g. furosemide) together with lithium without apparent problems.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 252-253.

The correct answer is: Furosemide

Question 16

Not answered

Marked out of 1.00

The safest diuretic to be used with lithium is

Select one:

- Bendrofluomethiazide
- Furosemide
- Xipamide
- Chlorthalidone
- Indapamide

Your answer is incorrect.

The following are loop diuretics- furosemide, bumetanide and torasemide. Although there are case reports of lithium toxicity induced by loop diuretics, many patients receive this combination of drugs without apparent problems. Patients taking blood diuretics may also have been advised to restrict their salt intake, which may contribute to the risk of lithium toxicity in these individuals. With the use of Thiazide diuretics (bendroflumethiazide, chlorthalidone, indapamide, metolazone, xipamide), care is required, as these classes of drugs can reduce the renal clearance of lithium and precipitate toxicity. (Ref: The Maudsley prescribing guidelines)

The correct answer is: Furosemide

Question 17

Not answered

Marked out of 1.00

A patient on MAOI can consume

Select one:

- Grapefruit juice
- Broad beans
- Red wine
- Pickled herring
- Cheese

Your answer is incorrect.

Explanation:

The most worrisome side effect of MAOIs is the tyramine-induced hypertensive crisis ('cheese reaction'). The amino acid tyramine is normally transformed via GI metabolism. However, MAOIs inactivate GI metabolism of dietary tyramine, thus allowing intact tyramine to enter the circulation. A hypertensive crisis may subsequently occur because of a powerful pressor effect of the amino acid. Sources of dietary tyramine include cheese, meat extracts and yeast extracts, alcohol, non-fresh fish, non-fresh poultry, offal, avocado, banana skins, broad bean pods, caviar, and herring (pickled or smoked). Unlike the others listed, grapefruit juice is not rich in tyramine.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 645.

Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 282.

The correct answer is: Grapefruit juice

Question 18

Not answered

Marked out of 1.00

Which of the following is true regarding diazepam?

Select one:

- Has 95% protein binding
- Has an inactive metabolite called desmethyldiazepam
- Does not cause blurred vision
- Half-life is 12 hours
- Presence of food increases the extent of absorption

Your answer is incorrect.

Explanation:

Diazepam has an elimination half-life of 20-50 hours. The fact that all benzodiazepines are lipid-soluble to varying degrees means that benzodiazepines and their active metabolites bind to plasma proteins. The extent of this binding is proportional to their lipid solubility. The amount of protein binding varies from 70-99% (approximately 95% for diazepam). Diazepam is oxidised to desmethyldiazepam, which, in turn, is hydroxylated to produce oxazepam. These products undergo glucuronidation to inactive metabolites. Food does not affect absorption.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 231.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 670.

The correct answer is: Has 95% protein binding

Question 19

Not answered

Marked out of 1.00

Which one among the following statements about mianserin is true?

Select one:

- Inhibits uptake of both Noradrenaline and 5-HT
- Has considerable alpha 1 adrenoreceptor blocking properties
- Has a plasma half life that allows once daily dosage
- Has strong antimuscarinic effects
- Has significant sedative effects

Your answer is incorrect.

Mianserin is a potent inhibitor of Noradrenaline uptake with little effect on 5-HT. It is an alpha 2 antagonist and has weak alpha 1 antagonism. It is mildly sedative and has weak antimuscarinic effects. It is rapidly absorbed from the gut with the peak plasma concentration appearing in less than 3 hours after a single oral dose. The plasma half-life is 10-29 hours.

The correct answer is: Has a plasma half life that allows once daily dosage

Question 20

Not answered

Marked out of 1.00

Which of the following is NOT a phase 2 metabolic reaction in drug elimination?

Select one:

- N acetylation
- Conjugation
- Methylation
- Hydrolysis
- Glucuronidation

Your answer is incorrect.

Conjugation, N-acetylation, methylation and glucuronidation are phase 2 reactions. Oxidation and hydrolysis are phase 1 reactions.

The correct answer is: Hydrolysis

Question 21

Not answered

Marked out of 1.00

Which of the following antidepressants' plasma levels correlates with therapeutic response?

Select one:

- Imipramine
- Mirtazapine
- Amoxapine
- Sertraline
- Duloxetine

Your answer is incorrect.

The clinical outcome when using imipramine is unequivocally related to the plasma level. Patients with levels less than 180 ng/ml had a low probability of recovery while levels above 180 ng/ml were associated with a high probability of recovery. Unlike nortriptyline, there is no tight therapeutic window for the efficacy of imipramine.

The correct answer is: Imipramine

Question 22

Not answered

Marked out of 1.00

Which of the following truly reflects the effects of smoking?

Select one:

- increases clozapine levels
- no tolerance develops to physical effects
- inhibits CYP1A2
- increases caffeine metabolism
- reduces risk of depression in adults

Your answer is incorrect.

Explanation:

Tobacco smoke contains polycyclic aromatic hydrocarbons that induce (increase the activity of) certain hepatic enzymes (CYP1A2 in particular). The extent of enzyme induction is determined by the number and type of cigarettes smoked and the degree of smoke inhalation. For some drugs used in psychiatry, smoking significantly reduces drug plasma levels and higher doses are required than in non-smokers. Smoking reduces plasma clozapine levels by up to 50%. Cigarette smoke increases caffeine metabolism and smokers may require higher doses of caffeine to gain desired effects.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 856, 867.

The correct answer is: increases caffeine metabolism

Question 23

Not answered

Marked out of 2.00

Neurohistology

Match the following neuroglial cells with their specific functions from the list provided:

Astrocytes Choose...

Ependymal cells Choose...

Neuronal support in sensory and autonomic ganglia
Act as scavenger cells at sites of CNS injury
Nutrition of neuronal cells
CNS myelin sheath formation
Neurilemma formation
Apoptotic clearance
Peripheral myelin sheath formation
Covering the ventricles

Your answer is incorrect.

Explanation:

Astrocytes are the most numerous of all three types of glial cells. These are star-shaped cells that enable nutrition of neurons, breakdown of some neurotransmitters, and maintaining the blood-brain barrier.

Ependymal cells are a particular type of glia that cover the ventricles and enable CSF circulation. They are any of the neuroglial cells of the ependyma in the ventricles of the brain, with cilia on their surfaces helping to propel cerebrospinal fluid through the ventricular system. Specialised ependymal cells form the choroid plexus.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 924.

Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015. p. 250.

The correct answer is: Astrocytes → Nutrition of neuronal cells, Ependymal cells → Covering the ventricles

Question 24

Not answered

Marked out of 3.00

Adverse effects

For each of the following drugs, identify the most important side effect associated with its therapeutic use

Galantamine	Choose...
Venlafaxine	Choose...
Topiramate	Choose...

Weight Loss
Prolactin elevation
Hypersalivation
Blurred vision
Polydipsia
Glucose dysregulation
Bradycardia
Diastolic Hypertension
Amenorrhoea
Nasal Stuffiness
QTc Prolongation

Your answer is incorrect.

Antidementia drugs like galantamine can produce bradycardia, especially in those with supraventricular conduction delay.

Venlafaxine can cause secondary hypertension in psychiatric patients. A significant number of patients receiving doses above 300mg/day of venlafaxine experience an increase in diastolic blood pressure. This risk is not restricted to those with preexisting hypertension.

Topiramate is a weak inhibitor of renal carbonic anhydrase and can promote the development of renal stones. It is weight neutral and can even cause weight loss. Adverse effects of Topiramate include taste perversion, anorexia and weight loss.

The correct answer is: Galantamine → Bradycardia, Venlafaxine → Diastolic Hypertension, Topiramate → Weight loss

Question 25

Not answered

Marked out of 3.00

Adverse effects

For each of the adverse effects given below, identify the drug most likely associated with it:

Renal failure	Choose...
Seizures	Choose...
Nocturnal myoclonus	Choose...

Lamotrigine
Acamprosate
Chlorpromazine
Lithium
Bupropion
Mirtazapine
Droperidol
SSRIs
Carbamazepine
Lorazepam
Sidenafil

Your answer is incorrect.

Lithium: Renal damage may occur in severe, prolonged toxicity - but cumulative lithium use leads to renal failure more commonly. It takes at least ten years to develop interstitial fibrosis - the pathology behind the chronic renal damage.

SSRIs: Severe sweating especially nocturnally is seen in some patients; Terazosin is effective in counteracting sweating. Nocturnal myoclonus is reported with SSRIs. The repetitive leg movements are characterised by extensions of the large toe and flexion of all major leg joints (ankle, knee, and hips) and occur every 20 to 60 seconds. Benzodiazepines and levodopa may be tried.

Bupropion is associated with seizures and exacerbation of eating disorders due its anorexigenic properties.

The correct answer is: Renal failure → Lithium, Seizures → Bupropion, Nocturnal myoclonus → SSRIs

Question 26

Not answered

Marked out of 4.00

Development of Adolescent Identity.
Match the stages of adolescent identity development based on Marcia's structural interview

Identity Achievement
Identity Moratorium
Identity Confusion
Identity Foreclosure
Identity Theft
Identity Diffusion

An 18-year-old man is in the process of developing a sense of identity when he leaves home to join the military. Here he has the opportunity to learn and develop new skills, which later on in life are incorporated into his sense of self.

A 17-year-old woman spends most of her time outdoors, in the company of a mixed group of young people. They spend time drinking alcohol, smoking cannabis; they have no plans for the future. Choose...

A 17-year-old man is given the choice by his parents to stop attending church if he wishes to. He doesn't wish to make an exploration of his own, but continues to attend church with his family, and to conform to their way of practicing religion. Choose...

A 21-year-old man after a lengthy period of exploration and self-searching, feels that he has a steady and stable sense of self. He finds himself committed to his work, with a clear sense of his goals and beliefs. Choose...

Your answer is incorrect.

Marcia distinguished four identity statuses based on the amount of exploration and commitment the adolescent experiences or has experienced

1) Identity Moratorium: This is a stage in which identity achievement is postponed whilst an intense exploration, which may be experienced as a crisis. At this point the young person is not yet ready to make a firm commitment.

2) Identity Diffusion: Identity issues have not been considered in detail and no firm commitments have been made for the future. During this stage the individual does not commit to any goals; it is characterized by procrastination, avoidance of responsibility and is the opposite of identity achievement.

3) Identity Foreclosure: Identity issues have not been considered seriously but future commitments have been made in spite of this. In this stage teenagers accept and adopt the traditional values and cultural norms of their parents or surrounding culture, rather than exploring and forming their own.

4) Identity Achievement: Various alternatives have been carefully considered and firm future commitments have been made. In this stage the individual is said to have solved their identity crisis and is able to make commitments to goals, values and beliefs.

(Ref:https://moodle.gllm.ac.uk/pluginfile.php/94785/mod_resource/content/1/Amanda_Tillson/Unit_4/Book_chapter_Adolesence_to_old_age.pdf)

The correct answer is: An 18-year-old man is in the process of developing a sense of identity when he leaves home to join the military. Here he has the opportunity to learn and develop new skills, which later on in life are incorporated into his sense of self. → Identity Moratorium, A 17-year-old woman spends most of her time outdoors, in the company of a mixed group of young people. They spend time drinking alcohol, smoking cannabis; they have no plans for the future. → Identity Diffusion, A 17-year-old man is given the choice by his parents to stop attending church if he wishes to. He doesn't wish to make an exploration of his own, but continues to attend church with his family, and to conform to their way of practicing religion. → Identity Foreclosure, A 21-year-old man after a lengthy period of exploration and self-searching, feels that he has a steady and stable sense of self. He finds himself committed to his work, with a clear sense of his goals and beliefs. → Identity Achievement

Question 27

Not answered

Marked out of 3.00

Developmental milestones (Language)
Identify the age at which children should reach the following milestones in language development (3 questions)

1. Develops intonation and babbles as though speaking, but does not say words. ✖

2. Uses 3-4 words together, and can change the order of words to make questions. ✖

3. Understands puns and jokes. ✖

0-6 months
6-12 months
12-18 months
18-24 months
2-3 years
4-5 years
6-7 years
8-9 years

Explanation:

1) 6-12 months: Babbling (consonant-vowel combinations) and responds when spoken to. 2) 2-3 years: By 3 years, children start using 3 to 4 word sentences with verbs that are generally understood by all adults. At this stage the 'wh' are common (who, why, where) 3) 6-7 years: Understands common idioms. Understands puns, humour

The correct answer is:

Develops intonation and babbling: 6-12 months

Uses 3-4 words together: 2-3 years

Understands puns and jokes: 6-7 years

Question 28

Not answered

Marked out of 5.00

Personality disorders
What is the most likely diagnosis for each of the following patients?

Antisocial personality disorder
Schizotypal personality disorder
Dependent personality disorder
Histrionic personality disorder
Schizoid personality disorder
Boderline personality disorder
Narcissistic personality disorder

Tod is a 25-year-old male who works in a family-run cornershop. He forgot to lock-up after work, which resulted in the store being burgled and a financial loss for the business. Tod appeared to demonstrate no concern for his actions. This is not the first

Choose...

Sabine is a 19-year-old fashion student. She holds the self-perception of being unique and superior to her peers. She told her supervisor that she felt she was exceptional and that was going to be the 'next Chanel', despite scoring a satisfactory grading

Choose...

Seema is a 33-year-old doctor who irritates her friends by speaking over them at dinner parties. Her friends complain that she 'always has to be the centre of attention.'

Choose...

Jason is a 45-year-old courier. He has a poor rapport with his work colleagues and tends to withdraw socially. Despite working for his company for fifteen years, he has never attended an out-of-work social gathering. Jason's colleagues suspect that he is into some cult worship and carries unusual ideas.

Choose...

21-year-old history student, Raphael, met up with his dissertation supervisor, who was baffled to learn that Raphael did not appear to be remotely bothered with the feedback he received.

Your answer is incorrect.

1) In Antisocial personality disorder, individuals show a lack of concern for others, irresponsibility, irritability, aggression, inability to maintain enduring relationships and disregard and violation of others' rights

2) In narcissistic personality disorder, grandiosity, lack of empathy and the need for admiration are common traits

3) In histrionic personality disorder, individuals show a high-degree of self-dramatization, shallow affect, egocentricity, craving attention, excitement and manipulative behaviour

4) Interpersonal discomfort with peculiar ideas, perceptions, appearance and behaviour are common traits in schizotypal personality disorder

5) Emotionally cold attitude, detachment, lack of interest in others, indifference to praise or criticism, excessive introspection and fantasy are common traits of schizoid personality disorder.

The correct answer is: Tod is a 25-year-old male who works in a family-run cornershop. He forgot to lock-up after work, which resulted in the store being burgled and a financial loss for the business. Tod appeared to demonstrate no concern for his actions. This is not the first → Antisocial personality disorder, Sabine is a 19-year-old fashion student. She holds the self-perception of being unique and superior to her peers. She told her supervisor that she felt she was exceptional and that was going to be the 'next Chanel', despite scoring a satisfactory grading → Narcissistic personality disorder, Seema is a 33-year-old doctor who irritates her friends by speaking over them at dinner parties. Her friends complain that she 'always has to be the centre of attention.' → Histrionic personality disorder, Jason is a 45-year-old courier. He has a poor rapport with his work colleagues and tends to withdraw socially. Despite working for his company for fifteen years, he has never attended an out-of-work social gathering. Jason's colleagues suspect that he is into some cult worship and carries unusual ideas. → Schizotypal personality disorder, 21-year-old history student, Raphael, met up with his dissertation supervisor, who was baffled to learn that Raphael did not appear to be remotely bothered with the feedback he received. → Schizoid personality disorder

Question 29

Not answered

Marked out of 5.00

Sleep disturbance

Which of the following best describes the following situation?

Susie is a 27-year-old nurse. She has a three-month-old son who cries during the nighttime and, as a result, has poor sleep. She complains of feeling excessively tired during the day.

George is a 53-year-old male who was recently prescribed aripriprazole for his schizophrenia. His wife complains that he can't keep his legs still at night and that he tends to kick her in bed

Shilpa, an 8-year-old girl, wakes up repeatedly in the night, fearful and feeling that she is going to die

Juliana is a 46-year-old lady who suffers from acute episodes of collapse on laughter.

You meet Vanessa, a four-year-old girl, in your CAMHS outpatient clinic. Her mother tells you that Vanessa regularly wakes up distressed, complaining that she has had scary dreams.

Akathisia
REM sleep disorder
Hypersomnolence
Nightmares
Cataplexy
Non-organic sleep disturbance
Sleep terrors

Choose...

Choose...

Choose...

Your answer is incorrect.

1) Hypersomnolence: recurrent episodes of excessive daytime sleepiness. It can be caused by chronic sleep deprivation, whereby an individual fails to get sufficient sleep for at least three months. This case is an example of secondary hypersomnia.
2) Akathisia is a movement disorder characterized by restlessness and an urge to be in constant motion. Behaviourally, this manifests as rocking while sitting or standing, marching on the spot, and the crossing and uncrossing of legs whilst sitting down
3) During sleep terrors, there are high levels of autonomic arousal several times at night. It is a sleep disorder in which the individual experiences feelings of terror or dread.
4) This is a case of cataplexy, a condition whereby strong emotions or laughter cause a person to suffer sudden physical collapse. The patient remains conscious. Low orexin levels can be one of the common risk factors.
5) A nightmare is an unpleasant, 'bad dream' that causes a strong emotional response.

The correct answer is: Susie is a 27-year-old nurse. She has a three-month-old son who cries during the nighttime and, as a result, has poor sleep. She complains of feeling excessively tired during the day. → Hypersomnolence, George is a 53-year-old male who was recently prescribed aripriprazole for his schizophrenia. His wife complains that he can't keep his legs still at night and that he tends to kick her in bed → Akathisia, Shilpa, an 8-year-old girl, wakes up repeatedly in the night, fearful and feeling that she is going to die → Sleep terrors, Juliana is a 46-year-old lady who suffers from acute episodes of collapse on laughter. → Cataplexy, You meet Vanessa, a four-year-old girl, in your CAMHS outpatient clinic. Her mother tells you that Vanessa regularly wakes up distressed, complaining that she has had scary dreams. → Nightmares

Question 30

Not answered

Marked out of 5.00

Measurement tools used in older subjects

Choose the most appropriate tool for each of the following situations in older subjects with cognitive impairment;

1. Monitoring cognitive decline (choose 2 options)

2. Monitoring decline in function of ADLs

3. Monitoring depression (choose 2 options)

Abbreviated Mental Test Score (AMTS)
MOCA
MADRS
CAMCOG
Hamilton Rating Scale
Cloc-drawing test
BEHAVE-AD
FAB

Explanation:

1) There are many tools for measuring aspects of cognition, function and mental state in Alzheimer's disease. Though not the most sensitive/specific, the most commonly used tool to measure cognitive decline is the MMSE, however, that is not an option here. The MOCA and CAMCOG are two appropriate tools to assess cognition. The AMTS would not be sensitive enough to pick up cognitive decline, and the clock-drawing test and FAB, though assessing some aspects, are not general enough. 2) For assessment of activities of daily living, the OT department have many tools. The association don't endorse one specific tool above all in Alzheimer's but note that some aspects of each tool may need to be modified. From this list, the MOHOST is the only assessment of ADLs included. The BEHAVE-AD is an assessment of behaviour in Alzheimer's disease. 3) The MADRS and Hamilton Rating scales are both used to assess for depression in the elderly.

The correct answer is:

Monitoring cognitive decline: MOCA and CAMCOG

Monitoring decline in function of ADLs: MOHOST

Monitoring depression: Hamilton Rating Scale and MADRS

Question 31

Not answered

Marked out of 1.00

A 24-year-old man says 'My nurse needlebottoms me weeks'. He is exhibiting

Select one:

- Paragrammatism
- Parapraxis
- Tangentiality
- Neologism
- Vorbeireden

Your answer is incorrect.

Explanation:

Paragrammatism refers to constructing sentences where words are jumbled giving rise to grammatically incorrect, but still meaningful sentences. It is the misuse of grammar not attributable to an overall reduction of grammatical morphemes or simplification of syntactic structure. Paragrammatic patients have prodigious output, leading to 'confused sentence monsters'. Examples are: *'And I want everything to be so talk.' 'She was handled to look at the books a bit.'*

Ref: Matchin W et al. Agrammatism and paragrammatism: a cortical double dissociation revealed by lesion-symptom mapping. Neurobiology of Language. 2020 Jun 1;1(2):208-25.

The correct answer is: Paragrammatism

Question 32

Not answered

Marked out of 1.00

During ward rounds, a psychotic patient was enquired about his appetite and dietary intake. He described a good appetite but referred to his stomach as a 'Food vessel'. What kind of speech disorder is this?

Select one:

- Echolalia
- Paraphasia
- Neologisms
- Perseveration
- Verbigeration

Your answer is incorrect.

Paraphasia are called as 'word approximations'. These are normal words that are used in an unconventional or distorted way but the derivation can be understood, even if bizarre. Neologisms are newly formed words or phrases whose derivation cannot be understood. These are created to express a concept for which the subject has no dictionary word.

The correct answer is: Paraphasia

Question 33

Not answered

Marked out of 1.00

An elderly woman believes that her neighbours enter her flat at night through the bathroom wall. This phenomenon is known as

Select one:

- Extracampine hallucination
- Doppelganger effect
- Capgras delusion
- Functional hallucination
- Partition delusion

Your answer is incorrect.

Explanation:

Partition delusions have been defined as 'the belief that people, animals, materials, or radiation can pass through a structure that would normally constitute a barrier to such passage. They arise as a secondary phenomenon to the primary delusional experience of being observed, spoken about, or physically affected by some agent outside the home.' This type of delusion seems to be particularly common in older adults who develop psychosis.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 551.

The correct answer is: Partition delusion

Question 34

Not answered

Marked out of 1.00

The most common type of delusion in delusional misidentification is

Select one:

- Jealousy
- Love
- Persecutory
- Reference
- Grandiose

Your answer is incorrect.

Explanation:

Delusional misidentification is a delusional belief that certain individuals are not who they externally appear to be. These syndromes usually manifest as symptoms of an underlying disorder (e.g. schizophrenia, mood disorder, delusional disorder, organic disorder) and rarely occur in isolation. They commonly involve an element of persecution, though not always. Types of delusional misidentification syndromes include Capgras, Fregoli, Intermetamorphosis, Subjective doubles, Autoscopic, Reverse subjective double, and Reverse Fregoli.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 106, 240.

The correct answer is: Persecutory

Question 35

Not answered

Marked out of 1.00

For every question in MMSE, a patient keeps giving the same answer 'Newcastle upon Tyne', which was the right answer for the first question regarding the city where she lives. She is exhibiting

Select one:

- Perseveration
- Metonymy
- Stereotypy
- Logorrhea
- Logoclonia

Your answer is incorrect.

Perseveration is defined as the contextually inappropriate and unintentional repetition of a verbal response or a motor behaviour. It is associated with organic brain disease and is occasionally seen in Schizophrenia.

The correct answer is: Perseveration

Question 36

Not answered

Marked out of 1.00

A man saw a young woman in a pub and has repeated images of her all the time. He wonders why he gets these images so often and says that he never experienced any such thing before, even though it is enjoyable. This phenomenon is best termed as

Select one:

- Obsessive imagery
- Compulsions
- Eidetic imagery
- Flashback
- Preoccupation

Your answer is incorrect.

Pleasurable repetitions are unlikely to be obsessions

The correct answer is: Preoccupation

Question 37

Not answered

Marked out of 1.00

A 46-year-old man suddenly believes that his wife is unfaithful to him. He arrived at this conclusion following seeing a blue car. This is a

Select one:

- Secondary delusion
- Delusional mood
- Delusions of love
- Primary delusion
- Erotomania

Your answer is incorrect.

Explanation:

A delusional perception is a primary delusion which is recalled as having arisen as a result of a perception (e.g. a patient who, on seeing two white cars pull up in front of his house, became convinced that he was therefore about to be wrongly accused of being a paedophile). The percept is a real external object, not a hallucinatory experience.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 107.

The correct answer is: Primary delusion

Question 38

Not answered

Marked out of 1.00

Which of the following is true regarding sensory deprivation?

Select one:

- Achieved by rendering stimuli patternless and meaningless
- Achieved using translucent goggles
- Prolonged sensory deprivation causes perceptual abnormalities
- Achieved using continuous white noise
- Prolonged sensory deprivation causes a sensation of euphoria

Your answer is incorrect.

Sensory deprivation must be differentiated from perceptual deprivation. In the former no sensations are allowed to be experienced by the subject and is a state of being cut off from almost all sensory stimulation from the external environment; in the latter, sensations are experienced but in a form that is not conducive to form full perceptions e.g. continuous white noise, translucent goggles etc which render the stimuli patternless or meaningless. Prolonged sensory deprivation can cause perceptual errors and a sense of intense dysphoria (unpleasant feeling).

The correct answer is: Prolonged sensory deprivation causes perceptual abnormalities

Question 39

Not answered

Marked out of 1.00

Repeated grandiose lies of a patient who seems to believe in his own untruthful statements is best described as

Select one:

- paramnesia
- cryptamnesia
- pseudologia fantastica
- malingering
- confabulation

Your answer is incorrect.

Explanation:

Pseudologia fantastica is typified by these characteristics: (1) the stories are not entirely improbable and are often built upon a matrix of truth; (2) the stories are enduring; (3) the stories are not told for personal profit per se and have a self-aggrandizing quality; and (4) they are distinct from delusions in that the person, when confronted with facts, can acknowledge these falsehoods. Pseudologica fantastica is referred to by various names: 'pathological lying', 'mythomania', and 'deception syndrome', among others. It is significantly different than mere lying. It involves disproportionate falsifications that may be extensive and complicated, present over a period of years or a lifetime, with no external gain or motive. In other words, this lying serves no obvious purpose, appears impulsive, and may be associated with an element of 'dyscontrol'. It is often associated with antisocial personality disorder, narcissistic personality disorder, borderline personality disorder, histrionic personality disorder, and PTSD. However, it is most often associated with factitious disorder, which is conceptualised as the intentional production of physical or psychological signs or symptoms for no obvious external gain and without clear reasoning behind the behaviour.

Ref: Frierson RL, Joshi KG. Implications of pseudologia fantastica in criminal forensic evaluations: a review and case report. Journal of forensic sciences. 2018 May;63(3):976-9.

The correct answer is: pseudologia fantastica

Question 40

Not answered

Marked out of 1.00

In therapy, a man recalled seeing a red train when he was a child. He then added seeing his dad buying tickets on the other side of the platform. He reported that this was the last time he remembered seeing his dad. This phenomenon is called

Select one:

- Retrospective falsification
- Recollection
- Registration
- Confabulation
- Recognition

Your answer is incorrect.

Explanation:

The reintegration of a complete event from a variety of different components (or scenes) is called recollection. Recollection entails the extension of memory back to early childhood events, a time in the distant past. The actual reconstruction of these events comes through reminiscence, associations, and autobiographical linking of developmental events.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 745.

The correct answer is: Recollection

Question 41

Not answered

Marked out of 1.00

The fundamental psychopathology in a Capgras delusion is

Select one:

- Doppelganger
- Confabulation
- Dissociative amnesia
- Delusional perception
- Reduplicative paramnesia

Your answer is incorrect.

Capgras syndrome may be a form of reduplicative paramnesia (RP), a misidentification phenomenon where the individual believes in or experiences simultaneous duplication of a place, a person, or even one's self. Capgras and RP may have the same pathologic substrate (http://www.ncbi.nlm.nih.gov/pubmed/571979)

The correct answer is: Reduplicative paramnesia

Question 42

Not answered

Marked out of 1.00

A 43-year-old man presents with symptoms of mania associated with grandiose delusions. What type of delusions are the most common in mania?

Select one:

- Autochthonous delusion
- Primary delusion
- Secondary delusion
- Delusional mood
- Delusional memory

Your answer is incorrect.

Primary delusions are the direct result of psychopathology while secondary delusions can be understood as having arisen in response to other primary psychiatric conditions (E.g. a manic patient developing delusions of grandiosity, or a depressed patient developing delusions of poverty, a patient with dementia developing a delusion that people are stealing from their house and moving items, etc.)

The correct answer is: Secondary delusion

Question 43

Not answered

Marked out of 1.00

A very religious woman has periods of elated mood when she believes that she has obtained a Divine Gift that puts her on par with Mother Teresa. This can be described as a/an

Select one:

- Magical thinking
- Primary delusion
- Religious belief
- Secondary delusion
- Overvalued idea

Your answer is incorrect.

Explanation:

This is an example of secondary delusions of grandeur, which are unfounded beliefs of being invested with special status, a special purpose or special abilities, that arise in relation to a euphoric mood state. Secondary delusions can be understood as having arisen in response to other primary psychiatric conditions (e.g. a patient with severely depressed mood developing delusions of poverty or a patient with euphoric affect developing grandiose delusions).

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 106.

The correct answer is: Secondary delusion

Question 44

Not answered

Marked out of 1.00

"I have tingling feelings in my hands caused by magnetic currents from an alternator". Identify the psychopathological terms used to describe the following experience;

Select one:

- Thought broadcasting
- Delusions of reference
- Passivity of volition
- Somatic passivity
- Made impulse

Your answer is incorrect.

Explanation:

Somatic passivity is a first rank symptom. It consists of either somatic hallucinations or normal perceptions associated with the belief that an outside agent is responsible. Delusions of control are a group of delusions which are also known as passivity phenomenon or delusions of body passivity. They are considered first-rank symptoms of schizophrenia. The core feature is the delusional belief that one is no longer in sole control of one's own body. The individual delusions are that one is being forced by some external to feel emotions (passivity of affect or 'made affect'), to desire to do things (passivity of impulse or 'made impulse'), to perform actions (passivity of volition or 'made act'), or to experience bodily sensations (somatic passivity).

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 107.

The correct answer is: Somatic passivity

Question 45

Not answered

Marked out of 1.00

Identify the neurotic defence mechanism among the following options:

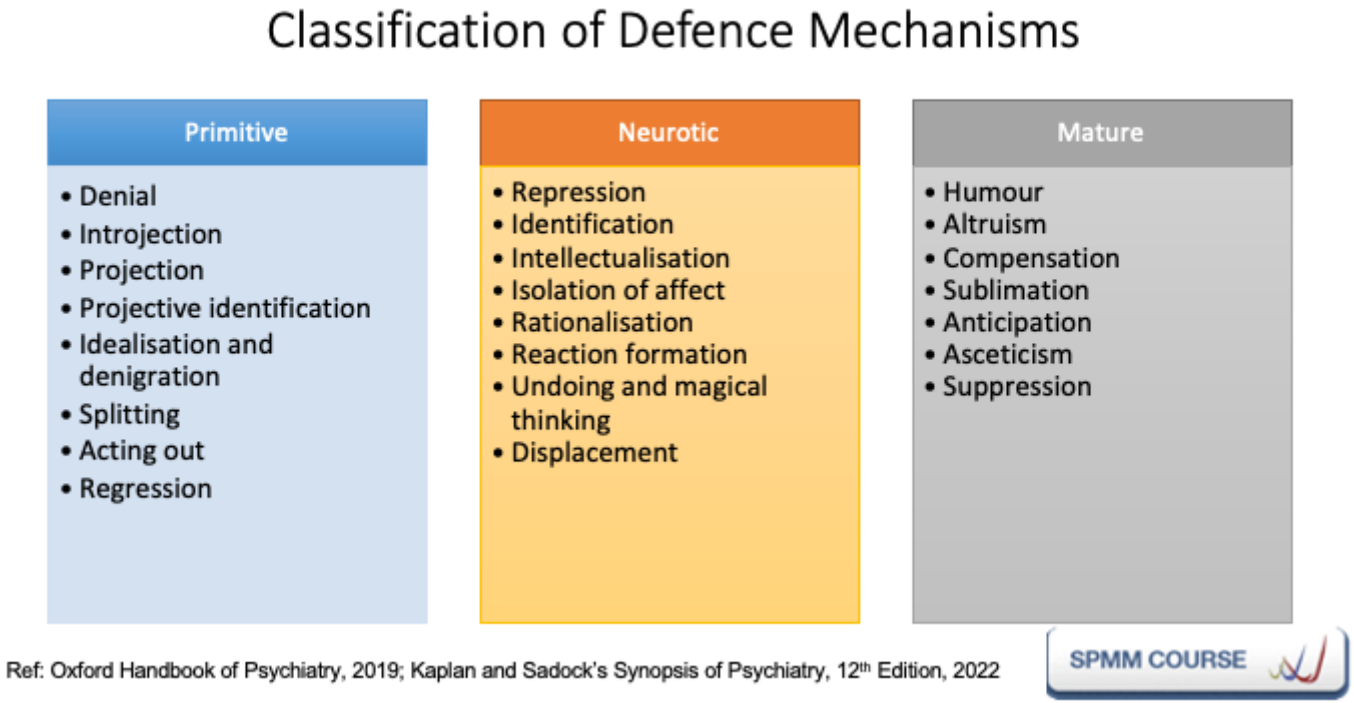
Select one:

- Acting out
- Sublimation
- Projection
- Intellectualisation
- Splitting

Your answer is incorrect.

Explanation:

Intellectualisation, repression, displacement, reaction formation, and dissociation are called neurotic defences.



Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. pp. 892-893.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 1066-1067.

The correct answer is: Intellectualisation

Question 46

Not answered

Marked out of 1.00

Which of the following defence mechanisms is thought to be related to the expression of paranoia?

Select one:

- undoing
- denial
- projection
- reaction formation
- isolation

Your answer is incorrect.

Explanation:

The psychoanalytic understanding of paranoia was initiated by Freud, who linked the defence mechanism of projection to the paranoid personality. Via projection, the paranoid person defends against unacceptable impulses, especially hate and aggression, which are also related to paranoid defiance. Projection is characterised as perceiving and reacting to unacceptable inner impulses and their derivations as though they were outside the self. On a psychotic level, this takes the form of frank delusions about external reality, usually persecutory, and includes both perceptions of one's own feelings and those of another with subsequent acting on the perception (psychotic paranoid delusions).

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 1066.

The correct answer is: projection

Question 47

Not answered

Marked out of 1.00

A 34 year old man who has a significant conflict with his partner has the following thoughts in the unconscious and the conscious level. "I hate him" (unconscious); "He hates me" (conscious). This refers to which of the following defence mechanisms?

Select one:

- Paranoia
- Reaction formation
- Denial
- Projection
- Sublimation

Your answer is incorrect.

Explanation:

Projection is the process through which unacceptable impulses are at once denied and attributed to someone else or to something in the environment. Thus, in projection, 'I hate him' is converted into 'he hates me.' Projection is a primitive or immature defense mechanism that involves attributing one's own internal unacceptable ideas and impulses to an external target, such as another individual, and reacting accordingly to them. It includes severe prejudice, rejection of intimacy through suspiciousness, hypervigilance to external danger, and injustice collecting.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 892.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022.

The correct answer is: Projection

Question 48

Not answered

Marked out of 1.00

The span of immediate memory is estimated to be around

Select one:

- 8±2
- 9±2
- 5±2
- 3±2
- 7±2

Your answer is incorrect.

Explanation:

Immediate memory refers to the amount of information that can be held in mind (like a phone number or digit span) to be available for immediate use. Its capacity is about seven items or chunks of information according to evidence first reviewed in an influential article by the psychologist George Miller in 1956. The 'magical number seven' (G. A. Miller; PA, Vol 31:2914,) demonstrated the concept of limited channel capacity and specified 7 (+/-2) as the immediate memory span.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022.

Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015. p. 694.

The correct answer is: 7±2

Question 49

Not answered

Marked out of 1.00

Which one of the following theories is associated with 'Retrieval failure of forgetting'?

Select one:

- State dependent memory
- Blockings
- Failure of prospective memory
- Context dependent memory
- All of the above

Your answer is incorrect.

All the above theories are associated with retrieval failure of forgetting and not due to encoding failure.

Encoding failure: Forgetting over time may occur due to defective or impaired initial encoding of information into memory. It may also be due to shallow processing of information.

Retrieval failure; This is the commonest cause of failure to access long term memory. Although the memory trace is there, the route to retrieving is blocked, which can be facilitated by cues.

The correct answer is: All of the above

Question 50

Not answered

Marked out of 1.00

Abnormalities in Stroop test suggests

Select one:

- Cortical blindness
- Lesion of optic chiasma
- Attentional bias
- Short term memory disturbance
- Damage to occipital cortex

Your answer is incorrect.

Explanation:

The Stroop effect is the delay and disruption in naming the colours of words printed in non-matching coloured ink, as when the word 'red' is printed in blue ink, the word 'blue' in green ink, and so on. To perform the task it is necessary to ignore the meanings of the printed words and to respond only to the colours in which they are printed, but with experienced adult readers automatic processing tends to occur, causing interference in the processing of information and a significant increase in the time required to name the colours, compared with the time taken to name the colours of meaningless strings of letters. Abnormal Stroop effect with the loss of appropriate inhibitory control is seen in frontal lobe lesions.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Attentional bias

Question 51

Not answered

Marked out of 1.00

Tom is a 4-year-old boy. He has a rabbit at home. When he reached out to stroke the rabbit, a hammer is struck against an iron rod causing a loud noise, which frightens him. This is repeated 5 times. As a result of this, the sight of the rabbit on its own now frightens Tom. This type of conditioning is called

Select one:

- Classical conditioning
- Operant conditioning
- Observational learning
- Stimulus discrimination
- Trace conditioning

Your answer is incorrect.

Explanation:

Classical conditioning was first described by Ivan Pavlov. It occurs when an initially neutral stimulus, such as ticking of a metronome, comes to elicit a particular response, such as salivation, as a consequence of being paired repeatedly with an unconditioned stimulus, such as food. In classical conditioning, the responses are often autonomic (e.g. salivation, change in heart rate).

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 1014.

Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015. p. 135.

The correct answer is: Classical conditioning

Question 52

Not answered

Marked out of 1.00

The process which leads to the transfer of information from short-term memory to long-term memory is called:

Select one:

- Rehearsal
- Encoding
- Consolidation
- Displacement
- Interference

Your answer is incorrect.

The transfer of information from short to long term, more permanent storage, is called memory consolidation. It is a process whereby the new information is fixed in long term memory and is integrated with existing stored information.

This can be facilitated by motivation (conscious effort) or association or elaboration of meaning.

Rehearsal or repetition also facilitates the transfer of items in short term memory to long term storage.

The correct answer is: Consolidation

Question 53

Not answered

Marked out of 1.00

All of the following are tests of recent memory except

Select one:

- Name and address recall task
- Rey's Memory test
- Three word learning task
- Wechsler memory scale
- Digit span

Your answer is incorrect.

Digit span is the commonest test of auditory, verbal working memory. Both forward and backward digit span are tested in routine clinical practice. The normal range of digit forwards is 6+/-1 and for reverse digit span is 5+/-1

Three word learning task (E.g. apple, table, penny) is a test of Anterograde memory and learning.

Name and address recall task (7 items) is the commonest test of recent memory. Here the subject is asked to recall as many items, without prompts, in five or ten minutes.

The correct answer is: Digit span

Question 54

Not answered

Marked out of 1.00

A middle-aged woman treated for agoraphobia has improved with repeated exposure to crowds. Which psychological process is involved here?

Select one:

- Instrumental conditioning
- Massed practices
- Cognitive restructuring
- Counter-conditioning
- Extinction

Your answer is incorrect.

Explanation:

Learned behaviour decreases if the unconditioned stimulus or reinforcer that was once paired with the conditioned stimulus or the response is eliminated from the situation. This phenomenon, called extinction, provides a means of eliminating unwanted behaviours learning through either form of conditioning.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 1015.

The correct answer is: Extinction

Question 55

Not answered

Marked out of 1.00

If a parent intentionally ignores a child's misbehaviour, this is an example of which operant conditioning principle?

Select one:

- Positive reinforcement
- Punishment
- Extinction.
- Negative reinforcement
- Shaping

Your answer is incorrect.

The attention given to a misbehaving child might positively reinforce the unwanted behaviour. By ignoring the misbehaving child, this positive reinforcement is withdrawn or extinguished (extinction). The phenomenon of extinction provided a means of eliminating unwanted behaviours learnt through either form of conditioning.

Negative reinforcement, on the other hand, is the removal of an undesirable effect that increases the probability of a target behaviour occurring in the future (e.g. flashing red lights are activated when a person starts a car without wearing seat belts. This is an undesirable effect. By buckling the seat belt, one can remove these aversive antecedents. Thus the target behaviour of 'buckling up' is reinforced by the 'removal' of the unwanted flashes of red light).

The correct answer is: Extinction.

Question 56

Not answered

Marked out of 1.00

Whose theory of emotions proposed that emotion is secondary to physiological changes and that there is a distinct pattern of changes for different emotions?

Select one:

- Cannon-bard theory
- Cognitive appraisal theory
- James Lange theory
- Schema theory
- None of the above

Your answer is incorrect.

According to James-Lange theory emotion, is secondary to physiological changes and that there is a distinct pattern of changes for different emotions.

The Cannon-Bard theory of emotion states that the subjective experience of emotions is independent of physiological changes. Any stimulus is processed by the thalamus that sends impulses to both the cortex and the hypothalamus. The cortex is responsible for emotions while the hypothalamus is responsible for physiological changes.

The cognitive appraisal conscious or unconscious always precedes emotional experience. Lazarus proposed this theory.

The correct answer is: James Lange theory

Question 57

Not answered

Marked out of 1.00

Who postulated that thinking occurs peripherally in the speech apparatus?

Select one:

- Sternberg
- John Watson
- Whorf
- Spearman
- Maslow

Your answer is incorrect.

The Linguistic relativity hypothesis also called as the Whorfian hypothesis, places language as primary and thoughts as secondary. But in peripheralism, John Watson postulated that thinking occurs peripherally in the speech apparatus.

The correct answer is: John Watson

Question 58

Not answered

Marked out of 1.00

Which of the following are correctly paired with the concepts they introduced?

Select one:

- Freud : negative and positive symptoms
- Jung : introversion
- Eugen Bleuler : dementia praecox
- Manfred Bleuler : dementia praecox
- Charcot : dissociation

Your answer is incorrect.

Jung developed a personality typology that distinguished between introversion and extroversion. He described 16 personality types including the differentiation between introverted and extroverted types.

Introverts are people who prefer their internal world of thoughts, feelings, fantasies, dreams, and so on, while extroverts prefer the external world of things and people and activities.

The correct answer is: Jung : introversion

Question 59

Not answered

Marked out of 1.00

Which one among the following is not one of the Gestalt principles of perception?

Select one:

- Proximity
- Similarity
- Closure
- Continuity
- Familiarity

Your answer is incorrect.

Gestalt law of perceptual organization includes Proximity: Objects close to each other are perceived as one figure, Closure: Incompletely closed figures are perceived as fully closed, Continuity: Continuous items are perceived together, Similarity: Similar items are grouped together based on colour or shape, etc. and Common fate: Things moving together are perceived together.

The correct answer is: Familiarity

Question 60

Not answered

Marked out of 1.00

Beautiful people are generally viewed as more kind, convincing and effective. This is called

Select one:

- Barnum effect
- Fundamental attribution error
- Halo effect
- Pygmalion effect
- Hawthorne effect

Your answer is incorrect.

Explanation:

Halo effect refers to our tendency to allow one specific trait or our overall impression of a person to influence our judgment of their other traits. In person perception, the halo effect is a generalisation from the perception of one prominent or salient characteristic, trait, or personality attribute to an impression of the personality as a whole, leading to inflated correlations between rated characteristics.

The Barnum effect is a tendency for people to accept vague, ambiguous, and generalised statements as being accurate descriptions of their own personalities.

Fundamental attribution error is a pervasive tendency to underestimate the importance of external situational pressures and to overestimate the importance of internal motives and dispositions in interpreting the behaviour of others.

The Pygmalion effect is a self-fulfilling prophecy whereby people tend to behave in the way others expect them to.

The Hawthorne effect is improvement in the performance or productivity of workers or students resulting from the introduction of new working methods or conditions, irrespective of the nature of the changes, the effect often being attributed to the feeling of being under concerned observation.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Halo effect

Question 61

Not answered

Marked out of 1.00

Which of the following ethical principles does the issue of capacity to make treatment decisions uphold?

Select one:

- Nonmaleficence
- Justice
- Confidentiality
- Autonomy
- Beneficence

Your answer is incorrect.

Explanation:

Individual freedom is based on the principle of autonomy in modern medical ethics. It is the principle that a person should be free to make his or her own decisions. Autonomy requires that a person act intentionally after being given sufficient information and time to understand the benefits, risks, and costs of all reasonable options. If a patient is not in a state of mind to make decisions for themselves, mechanisms for alternative decision-making should be considered.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 829.

The correct answer is: Autonomy

Question 62

Not answered

Marked out of 1.00

Under the principle of necessity which of the following can be done without patient's consent?

Select one:

- Covert administration of medication in a competent non consenting adult
- Detention of a voluntary psychiatric inpatient who is attempting to leave hospital
- Sterilisation of a patient with repeated postpartum mania
- Liver transplantation to a patient who consumed overdose of paracetamol
- Blood sample drawn to confirm suspected myocardial infarction in a schizophrenia patient who is unconscious

Your answer is incorrect.

Emergency interventions for life-saving measures could be carried out under common law or necessity principle.

Irreversible or long term procedures cannot be carried out without formalised legal procedures.

The correct answer is: Blood sample drawn to confirm suspected myocardial infarction in a schizophrenia patient who is unconscious

Question 63

Not answered

Marked out of 1.00

Tarasoff case is related to which of the following ethical principles?

Select one:

- Autonomy
- Beneficence
- Confidentiality
- Justice
- Capacity

Your answer is incorrect.

Explanation:

On 27 October 1969, Tatiana Tarasoff was killed by her ex-boyfriend (Mr. P), who had previously declared his intentions during an appointment with his psychotherapist at the University of California at Berkley. The therapist notified the campus police, both orally and in writing, that the patient was dangerous and should be committed. Concerned about the breach of confidentiality, the therapist's supervisor vetoed the recommendation and ordered all records related to the patient's treatment destroyed. At the same time, the campus police temporarily detained Mr. P, but then released him. Two months later, Mr. P killed Tatiana. Her parents sued the university, and the Supreme Court ultimately decided that physicians and therapists have a duty to warn third parties of threatened danger arising from a patient's violent intentions. As a final statement, the Court stated that 'protective privilege ends where public peril begins.'

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 970.

Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 839.

The correct answer is: Confidentiality

Question 64

Not answered

Marked out of 1.00

According to the Holmes and Rahe Social readjustment scale, which of the following has the highest 'Life Change' value?

Select one:

- Pregnancy
- Retirement
- Death of a spouse
- Marriage
- Minor violation of law

Your answer is incorrect.

Explanation:

In 1967, psychiatrists Thomas Holmes and Richard Rahe examined over 5000 patients' medical records to determine whether stressful events cause illnesses. Patients ranked a list of 43 life events based on a relative score. Each event had a different 'weight' for stress. Their results were published as the Social Readjustment Rating Scale, now commonly known as the Holmes and Rahe Stress Scale. The scale lists common stressful events and arbitrarily assigns a value of 50 Life Change Units (LCUs) to the stress caused by marriage. Death of a spouse has the highest LCUs at 100, followed by divorce (73 LCUs), and marital separation (65 LCUs).

Ref: Noone PA. The Holmes–Rahe Stress Inventory. Occupational Medicine. 2017 Oct 1;67(7):581-2.

Holmes TH, Rahe RH. The social readjustment rating scale. Journal of psychosomatic research. 1967.

The correct answer is: Death of a spouse

Question 65

Not answered

Marked out of 1.00

Which of the following ethical theories concerns itself with the intent behind the actions of an individual, rather than the consequences?

Select one:

- Deontology
- Teleology
- Consequentialism
- Utilitarianism
- Virtue theory

Your answer is incorrect.

Explanation:

According to Immanuel Kant, a German philosopher, deontology is an ethical approach centered on rules and professional duties. Unlike many ethical theories, deontology does not focus on the consequences of individual actions. Personal emotions behind actions also do not matter with Kantian deontology because Kant believed humans do not always have rational control over their feelings. Instead, the intent behind chosen actions holds far more importance. Therefore, deontology proponents judge actions based on what people consider to be morally correct, regardless of actual consequences.

The correct answer is: Deontology

Question 66

Not answered

Marked out of 1.00

A patient demands to have a CT scan of her head to evaluate her migrainous headaches. Your clinical judgment suggests that the pain is due to a migraine and there is no necessity for a CT scan.She does not agree and repeatedly asks for the scan. The best option in this situation is

Select one:

- Do not order the CT scan and ask her to seek medical care elsewhere
- Do not order the CT scan, but discharge her from your care
- Order the CT scan so that the patient will not lodge a complaint with hospital managers.
- Do not order the CT scan, but offer to refer her for a second opinion if she wishes to consider this.
- Ask her to arrange for a privately paid CT scan if she is willing to pay for this.

Your answer is incorrect.

Ordering unnecessary tests is both inappropriate utilization of health care resources and needlessly expensive. The CT scan should not be ordered for this patient if it is not indicated, regardless of their request.

Provided the professional adheres to an existing level of standard of care to this patient, then a malpractice or negligence should not be of concern.

There is not a clear reason to discharge the patient from the caseload completely and to do so may be construed as patient abandonment.

The best choice, in this case, is to offer the patient an opportunity to get a second opinion on her symptoms.

The correct answer is: Do not order the CT scan, but offer to refer her for a second opinion if she wishes to consider this.

Question 67

Not answered

Marked out of 1.00

In some families, contradictory parental demands on children lead to confusion and frustration, increasing the risk of emotional difficulties. Which of the following term best captures this situation?

Select one:

- Learned Helplessness
- Rapproachment
- Marital Schism
- Double bind
- Projective identification

Your answer is incorrect.

Explanation:

Double bind is an inescapable dilemma involving conflicting demands that allow no right or satisfactory response. An influential theory of the aetiology of schizophrenia was put forward by Gregory Bateson and colleagues in 1956, according to which schizophrenia is caused by parenting styles that create double binds for children, as when a mother complains to her son for not giving her a kiss but recoils physically whenever the child does kiss her. Empirical evidence has not been forthcoming in support of this theory.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Double bind

Question 68

Not answered

Marked out of 1.00

Which of the following is not a risk factor for ADHD?

Select one:

- Large size of the family
- Family history of ADHD
- Low birth weight
- Low educational attainment in parents
- Food additives

Your answer is incorrect.

Explanation:

There is no proven link between food additives and ADHD. ADHD and its common complications are associated with low social class, large family size, paternal antisocial personality, and maternal mental disorder. Some have suggested that food additives, colourings, preservatives, and sugar are possible contributing causes of hyperactive behaviour. However, studies have not confirmed these theories. No research has established artificial food colouring nor sugar as causes of ADHD.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 138.

The correct answer is: Food additives

Question 69

Not answered

Marked out of 1.00

All of the following factors commonly influence the basic features of health explanatory belief models except

Select one:

- history of good physical health
- age
- time-point and context of assessment
- educational status
- gender

Your answer is incorrect.

Kleinman's work in the development of a health explanatory belief model has led to an increased appreciation of the role of a patients' understanding of the causes of an illness when attempting to treat the illness.

Age, gender, cultural background, time and context of assessment and educational status can influence one's explanatory models. Ref: Kramer et al., West J Med. 2002 Sep; 176(4): 227-231.

The correct answer is: history of good physical health

Question 70

Not answered

Marked out of 1.00

At what age does theory of mind develop?

Select one:

- 4 years
- 9 years
- 5 years
- 2 years
- 2 months

Your answer is incorrect.

Explanation:

'Theory of mind' refers to people's intuitive understanding of their own and other people's minds or mental states, including beliefs and thoughts. It develops by degrees from a very early age in humans: by the age of 3 years children are developing a theory of mind but cannot yet understand that people's beliefs can be false. This fully develops around 4 years of age. People with autistic spectrum disorder are severely impaired in their ability to understand mental states and to appreciate how mental states govern behaviour.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: 4 years

Question 71

Not answered

Marked out of 1.00

Which one of the following infants do not attempt to interact with the mother when she is present, nor do they appear particularly upset when left with the stranger?

Select one:

- Disorganised infants
- Anxious/resistant infants
- Securely attached infants
- Anxious/avoidant infants
- All of the listed options

Your answer is incorrect.

Mary Ainsworth devised the 'strange situation' procedure with 12 to 18 month infants which categorised infant-parent relationships.

Anxious/avoidant infants do not attempt to interact with the mother when with her nor do they appear particularly upset when left with the stranger. When the mother returns, the infant may not make any attempt to get close to her. There is a lack of proximity or interaction during the reunion.

The correct answer is: Anxious/avoidant infants

Question 72

Not answered

Marked out of 1.00

Which of the following experiments is not correctly matched with Piaget's stages of cognitive development?

Select one:

- Classification of objects according to rules - concrete operational stage
- Hiding objects - sensorimotor stage
- Pendulum experiment - formal operational stage
- Mountain tasks - pre-operational stage
- Conservation tasks - sensorimotor stage

Your answer is incorrect.

Explanation:

In the sensorimotor stage (birth to 2 years), infants learn to differentiate themselves from the world. They also become able to maintain a mental image of an object, even when it is not visible (object permanence).

The lack of mastery of conservation in children at the pre-operational stage was demonstrated by Jean Piaget through a series of experiments. He also used the 'three mountains task' to demonstrate that pre-operational children comprehend the world only from their own point of view and demonstrate 'egocentrism'.

It is not until the stage of concrete operations that egocentric thought is replaced by operational thought. Children can now see things from someone else's perspective. Children also understand the concept of conservation, which is the ability to recognise that, although the shape of objects may change, the objects still maintain or conserve other characteristics that enable them to be recognised as the same.

Piaget and Inhelder demonstrated that children's understanding of the physical world changes with growth by using pendulum experiment. At the start of formal operational thinking, children can construct experiments to deduct facts. They can work out the factor that determines swinging duration as string length and not weight or speed, etc.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 1007-1009.

Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015. p. 237.

The correct answer is: Conservation tasks - sensorimotor stage

Question 73

Not answered

Marked out of 1.00

A 4-year-old boy is playing a hide-and-seek game. He 'hides' the sought object on his head. He assumes that since he cannot see the object, others cannot see it too. In Piaget's terms, this is best described as

Select one:

- Artificialism
- Egocentrism
- Propositional thought
- Precausal reasoning
- Circular reaction

Your answer is incorrect.

Egocentrism refers to the restricted ability to view the world from a single point of view at a preoperational developmental stage. This was demonstrated using the Mountains task where a child at this age group could not say what a person would see from another side of the desk when only one side of a mountain was visible from each view.

The correct answer is: Egocentrism

Question 74

Not answered

Marked out of 1.00

Which one among the following fears is present at birth?

Select one:

- Fear of loud noises
- Fear of strangers
- Fear of animals
- Fear of dark
- Fear of supernatural things

Your answer is incorrect.

At birth, the newborn infant shows an innate reflex to sudden or loud noises. They also show a similar reaction to a sense of falling or sudden loss of physical support and cold blasts of air.

The correct answer is: Fear of loud noises

Question 75

Not answered

Marked out of 1.00

Family instability affects development of children at home. Which of the following is correct in this regard?

Select one:

- There is no effect on cognitive development
- Boys are less affected than girls
- Hyperactive children are more affected than easy-going children
- Younger children are less affected than older children
- Psychosis is the most common effect

Your answer is incorrect.

Family instability can affect children to various extent depending on sex (boys affected more than girls), age (younger affected more than older children), and temperament (hyperactive affected more than placid). This has demonstrable effect on a child's cognitive achievements; the most common psychopathology noted is behavioural difficulty.

The correct answer is: Hyperactive children are more affected than easy-going children

Question 76

Not answered

Marked out of 1.00

Which of the following is NOT an effect of parental divorce?

Select one:

- Guilt and accepting responsibility for separation
- Suicidal ideas
- Physical aggression
- Increased motivation for academic performance
- Changes in bowel habits

Your answer is incorrect.

Explanation:

A child's age at the time of the divorce affects the child's reaction to the divorce. Immediately after a divorce, an increase in behavioural and emotional disorders appears in all age groups. Infants do not understand anything about separation or divorce. However, they do notice changes in their parents' responses to them and may experience changes in their eating or sleeping patterns; have bowel problems; and seem more fretful, fearful, or anxious. Children 3-6 years of age may not understand what is happening, and those who do understand often assume that they are somehow responsible for the divorce. Older children, especially adolescents, comprehend the situation and may believe that they could have prevented the divorce had they intervened in some way, but they are still hurt, angry, and critical. Among boys, physical aggression is a common sign of distress. Suicide rates for children of divorce are much higher than for children from intact families.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 897.

The correct answer is: Increased motivation for academic performance

Question 77

Not answered

Marked out of 1.00

According to Erik Erikson, between the ages of 6 and 11, the crisis to be resolved is

Select one:

- Autonomy vs shame
- Initiative vs guilt
- Industry vs inferiority
- Trust vs mistrust
- Identity vs role confusion

Your answer is incorrect.

This is stage 4 of Erik Erikson's theory of psychosocial development. Its focus in 'Industry vs Inferiority'.

During 6 to 12 years of age we are capable of learning, creating and accomplishing numerous new skills and knowledge. The child develops 'industry' by learning new things and takes pride in the things made.

This is also a very social stage of development, and if we experience unresolved feelings of inadequacy and 'inferiority' among our peers, we can have serious problems in terms of competence and self-esteem.

The correct answer is: Industry vs inferiority

Question 78

Not answered

Marked out of 1.00

Which of the following programme is very appropriate for the needs of children and adults with communication and learning difficulties?

Select one:

- British Sign Language
- Makaton
- Sensory therapy
- Phonetic training
- Habit reversal

Your answer is incorrect.

The Makaton Vocabulary was originally devised in the 1970's by Margaret Walker a speech and language therapist and two of her colleagues Katharine and Tony, the name Makaton is made up of syllables from the beginning of each of their names Ma-Ka-Ton. Makaton consists of a core vocabulary of specially selected concepts and ideas. It contains approximately 350 signs and associated words.

These are considered to be most appropriate for the needs of children and adults with communication and language difficulties. It is specifically designed for people with learning difficulties of all kinds and their parents & carers.

The initial stages introduce the vocabulary required to express basic ideas. More complex concepts are introduced in the later stages. There is also an additional resource vocabulary grouped into topics.

The correct answer is: Makaton

Question 79

Not answered

Marked out of 1.00

Which of the following drugs can cause whitish-brown granular deposits in the cornea and lens ?

Select one:

- Thioridazine
- Amisulpride
- Chlorpromazine
- Quetiapine
- Olanzapine

Your answer is incorrect.

Explanation:

High-dose chlorpromazine can result in abnormal pigmentation of the eyelids, cornea, and lens. However, this is not associated with visual disturbance and resolves on cessation of treatment. Corneal oedema can also (rarely) occur with chlorpromazine, leading to loss of vision if treatment is not withdrawn. Thioridazine (now withdrawn) was associated with pigmentary retinopathy.

Ref: Taylor DM, Gaughran F, Pillinger T. The Maudsley Practice Guidelines for Physical Health Conditions in Psychiatry. 2021. p. 519.

The correct answer is: Chlorpromazine

Question 80

Not answered

Marked out of 1.00

Risk factors for tardive dyskinesia includes all of the following except

Select one:

- Using high potency drugs
- Having an affective disorder diagnosis
- Short term use of antipsychotics
- Concomitant anticholinergic treatment
- Having neurological damage

Your answer is incorrect.

Explanation:

Risk factors for tardive dyskinesia include chronic use of antipsychotics, change/cessation of chronic treatment, concomitant anticholinergic treatment, elderly (>60 years), female, organic disorder (e.g. dementia, intellectual disability, epilepsy), previous head injury, alcoholism, comorbid mood disorder, negative symptoms of schizophrenia, diabetes mellitus, history of previous drug-induced akathisia/Parkinsonism/dystonias, and concomitant use of predisposing drugs.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 1014.

The correct answer is: Short term use of antipsychotics

Question 81

Not answered

Marked out of 1.00

Rabbit syndrome presenting with mouth tremors is a result of

Select one:

- D1 receptor blockade
- 5HT2C receptor blockade
- D2 receptor blockade
- 5HT2A receptor blockade
- D4 receptor blockade

Your answer is incorrect.

The rabbit syndrome is a tremor affecting the lips and perioral muscles and occurs late in the course of antipsychotic treatment. It is considered as a form of extrapyramidal symptom related to D2 receptor blockade.

The correct answer is: D2 receptor blockade

Question 82

Not answered

Marked out of 1.00

A patient with schizophrenia has been functioning considerably better since he started taking clozapine. Currently he is on a dose of 600 mg/day, but he has experienced an episode of new onset seizures. Which of the following is the most appropriate next step?

Select one:

- Add carbamazepine to clozapine
- Add valproate to the same dose of clozapine
- Add amisulpride
- Decrease the dose of clozapine
- Stop clozapine for 6 weeks

Your answer is incorrect.

Seizures with clozapine are related to dose, plasma levels and rapid dose escalation.

In those patients who experience seizures on clozapine, prophylactic valproate can be considered if on a high dose or with high plasma level (more than 500 ng/l). After a seizure, withhold clozapine for one day; restart at reduced dose (half previous dose); give anticonvulsant like sodium valproate.

EEG abnormalities are common in those on clozapine and does not warrant alteration in the treatment automatically.

Ref: (The Maudsley Prescribing Guidelines 13th Ed)

The correct answer is: Decrease the dose of clozapine

Question 83

Not answered

Marked out of 1.00

Benzodiazepine withdrawal is associated with:

Select one:

- Increased brain GABA function
- Decreased brain dopamine function
- Decreased brain GABA function
- Increased brain serotonin function
- Increased brain dopamine function

Your answer is incorrect.

Benzodiazepines belong to a class of drugs that modulate the neurotransmitter GABA, particularly GABA A to open the chloride channel. They can modulate GABA A receptors as an agonist.

People become tolerant to benzodiazepines probably because their nerve cells respond by producing fewer receptors for GABA/benzodiazepines. This phenomenon, known as 'down-regulation', means that the number of 'high affinity' GABA receptors decreases in response to the enhancement of GABA caused by the drug. The body responds to the continued presence of the drug with a series of adjustments that tend to overcome the drug effects.

In the case of benzodiazepines, compensatory changes occur in the GABA and benzodiazepine receptors which become less responsive, so that the inhibitory actions of GABA and benzodiazepines are decreased (Ref: CH Ashton. The role of GABA and GABA receptors in benzodiazepine withdrawal. http://www.benzosupport.org/gaba.htm)

The correct answer is: Decreased brain GABA function

Question 84

Not answered

Marked out of 1.00

A 48-year-old gentleman suffers from schizophrenia and is on Olanzapine treatment. He has a BMI of 38 and suffers from drowsiness and erectile dysfunction. His diagnosis would most likely be;

Select one:

- Kluver bucy syndrome
- Cushing's syndrome
- Addison's disease
- Klein Levin syndrome
- Diabetes mellitus

Your answer is incorrect.

Erectile dysfunction (ED) is a common problem amongst men who have diabetes affecting 35-75% of male diabetics, which could be induced by Olanzapine. The other side effects of Olanzapine include drowsiness and weight gain. Antipsychotic induced sedation and weight gain may reduce sexual desire.

The correct answer is: Diabetes mellitus

Question 85

Not answered

Marked out of 1.00

Which of the following drugs can markedly increase blood acetaldehyde concentration?

Select one:

- Acamprosate
- Morphine
- Disulfiram
- Naltrexone
- Biperiden

Your answer is incorrect.

Aldehyde dehydrogenase (ALDH) is the enzyme that facilitates the catabolism of acetaldehyde to acetic acid. When the action of ALDH is inhibited by disulfiram, acetaldehyde is not converted to acetate but accumulates in the blood.

Most of the symptoms of the disulfiram-ethanol are due to the increased circulating acetaldehyde. It can lead to symptoms like flushing, headache, vomiting and palpitations if alcohol is consumed whilst taking Disulfiram.

The correct answer is: Disulfiram

Question 86

Not answered

Marked out of 1.00

A 33-year-old woman with a history of recurrent depressive disorder and is on fluoxetine 20 mg/day. She wishes to conceive and if she takes fluoxetine during pregnancy, what is the risk of it causing major malformations in the foetuses?

Select one:

- Trebles the risk
- Increases the risk by a small amount
- Doubles the risk
- Quadruples the risk
- More than quadruples the risk

Your answer is incorrect.

Explanation:

With respect to antidepressant use during pregnancy, **all SSRIs** are associated with small increased risks of congenital (primarily cardiac) malformations and, in late pregnancy, persistent pulmonary hypertension. Recent data on pregnancy outcomes with **antidepressant exposure** have shown no significant impact on rates of spontaneous abortion, a small impact on gestational age at delivery (<1 week earlier), a modest impact on Apgar scores, and a modest impact on birthweight (slightly lower, though not significantly different when compared with women with depression). **Paroxetine** is associated with a small increased risk of congenital, major congenital, and cardiac malformations. **Fluoxetine** is associated with a small increased risk of major congenital and cardiac malformations. **Citalopram and escitalopram** are associated with a small increased risk of ventral septal defects.

Ref: McAllister-Williams RH et al. British Association for Psychopharmacology consensus guidance on the use of psychotropic medication preconception, in pregnancy and postpartum 2017. Journal of Psychopharmacology. 2017 May;31(5):519-52.

National Institute for Health and Clinical Excellence. The NICE guideline on antenatal and postnatal mental health. Updated edition 2014.

The correct answer is: Increases the risk by a small amount

Question 87

Not answered

Marked out of 1.00

The drug of choice for treatment of restless legs syndrome is

Select one:

- Antidepressant
- Dopamine agonist
- Baclofen
- Clonazepam
- Antipsychotic

Your answer is incorrect.

Explanation:

Currently, dopamine agonists such as ropinirole, pramipexole, carbidopa/levodopa or pergolide are commonly used for the treatment of restless legs syndrome. Clonazepam may also be beneficial. Dopaminergic medications have been the mainstay treatment of restless legs syndrome (RLS). Doses of dopamine agonists used for RLS treatment are much lower than typical doses in patients with Parkinson disease and are timed to be taken approximately 2 hours before typical symptom onset.

Ref: Bollu PC et al. Sleep medicine: Restless legs syndrome. Missouri medicine. 2018 Jul;115(4):380.

Trotti LM. Restless legs syndrome and sleep-related movement disorders. Continuum: Lifelong Learning in Neurology. 2017 Aug 1;23(4):1005-16.

The correct answer is: Dopamine agonist

Question 88

Not answered

Marked out of 1.00

In patients receiving lithium therapy, the following can be observed even in normal therapeutic doses:

Select one:

- ECG changes
- Coarse tremors
- Bradycardia
- Pulmonary oedema
- Ataxia

Your answer is incorrect.

Lithium usually causes benign ECG changes (T wave changes, widening of the QRS complex).

Reversible flattening and inversion of T-waves are seen even at therapeutic doses, but usually produce no clinical consequences.

Br. J. Clin. Pharmac. (1980), 9, 599-604

The correct answer is: ECG changes

Question 89

Not answered

Marked out of 1.00

One among the following is not a feature of serotonin syndrome

Select one:

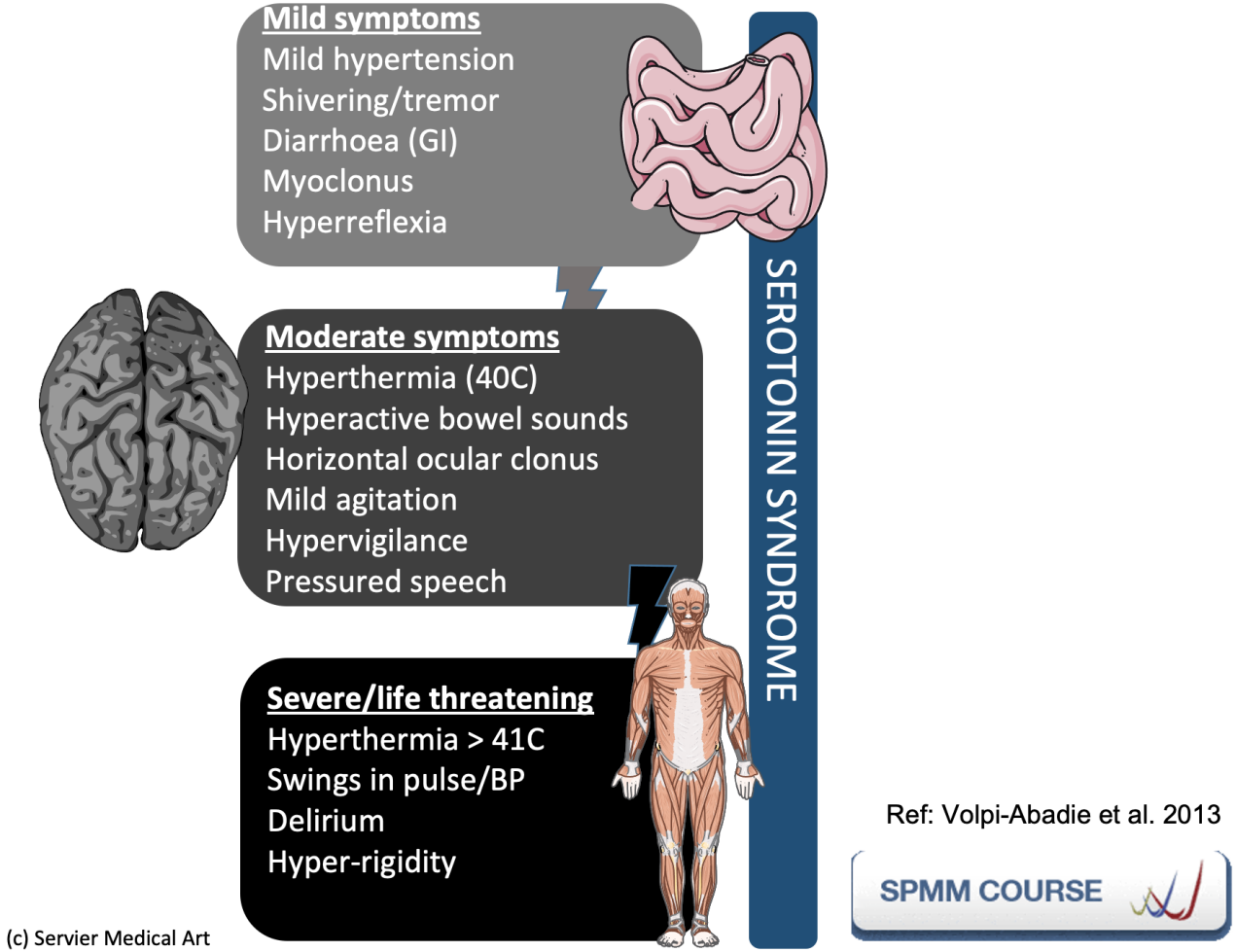
- Hyperreflexia
- Confusion
- Myoclonic Jerks
- Electric shock like sensations
- Autonomic instability

Your answer is incorrect.

Explanation:

Serotonin syndrome: Confusion, pyrexia, sweating, myoclonic jerks, hyperreflexia, autonomic instability, gastrointestinal symptoms, mood changes and mania. Note that shock-like sensations are seen in SSRI discontinuation syndrome.

Ref: Volpi-Abadie J et al. Serotonin syndrome. Ochsner Journal. 2013 Dec 21;13(4):533-40.



The correct answer is: Electric shock like sensations

Question 90

Not answered

Marked out of 1.00

Which of the following is a potentially serious adverse effect associated with tacrine?

Select one:

- Hepatotoxicity
- Blood dyscrasias
- Respiratory arrest
- Retinal damage
- Renal failure

Your answer is incorrect.

Tacrine which is the first cholinesterase inhibitor to be introduced has a potential for hepatotoxicity.

Elevated liver enzymes (potentially serious), anorexia, nausea, rash, myalgia, loss of weight, and increased sweating are some of the side effects of tacrine, the first introduced anti-dementia drug.

Donepezil is specific for cholinesterase inhibition in the CNS while tacrine is not. Donepezil, therefore, has less peripheral adverse effects.

The correct answer is: Hepatotoxicity

Question 91

Not answered

Marked out of 1.00

Which of the following is a predisposing factor for QT prolongation?

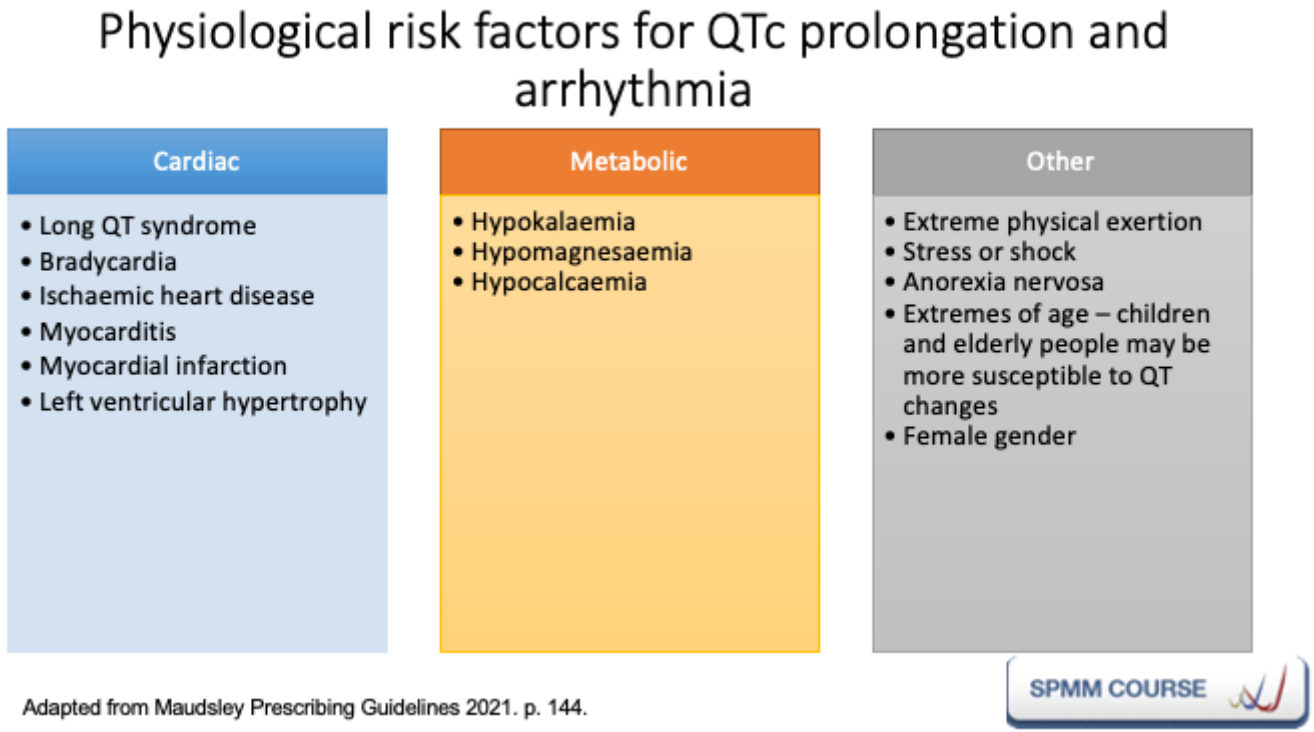
Select one:

- Hyperkalemia
- Hypermagnesemia
- Male gender
- Hypercalcemia
- Female gender

Your answer is incorrect.

Explanation:

New cases of long QT syndrome are diagnosed more in female patients (60-70% of cases) than male patients. The female predominance may be related to the relatively prolonged QTc (as determined by using the Bazett formula) in women compared with men and to a relatively higher mortality rate in young men. Electrolyte abnormalities (especially hypokalaemia or hypomagnesaemia) are also associated.



Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 144.

The correct answer is: Female gender

Question 92

Not answered

Marked out of 1.00

Thromboembolism as a possible side effect of clozapine treatment seems most likely to occur in the

Select one:

- First 6-8 weeks of treatment
- First 3 months of treatment
- First 4 days of treatment
- First 6 months of treatment
- First 4 weeks of treatment

Your answer is incorrect.

Explanation:

A possible association between clozapine and thromboembolism has been suggested. Thromboembolism may be related to clozapine's observed effects on antiphospholipid antibodies and platelet aggregation. It seems most likely to occur in the first 6 months of treatment but can occur at any time.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 222.

The correct answer is: First 6 months of treatment

Question 93

Not answered

Marked out of 1.00

Which of the following statements is true about neural correlates of sign language in deaf patients?

Select one:

- Occipital lobe takes over language processing functions in deaf patients
- Comprehension of sign language does not require involvement of language areas in brain
- Right sided dominance for language is seen in most sign language users
- Left temporal lesions can produce comprehension deficits in sign language users
- Aphasias are not seen in sign language users

Your answer is incorrect.

Deaf patients show a neural organisation of language cortex that is similiar to the non-deaf population. Left temporal cortex is important for comprehending sign language while Broca's area serves in the production of sign language.

The correct answer is: Left temporal lesions can produce comprehension deficits in sign language users

Question 94

Not answered

Marked out of 1.00

Damage to which of the following regions unilaterally will produce an upper quadrantanopia?

Select one:

- Superior frontal gyrus
- Medial temporal lobe
- Inferior Parietal Lobule
- Lower bank of calcarine fissure
- Parietal operculum

Your answer is incorrect.

The primary visual cortex lies on the banks of the calcarine fissure. Neurons located on the lower bank are connected to the lateral geniculate nucleus carrying visual information from the nasal or temporal upper visual fields. A lesion of this region can induce an upper quadrantanopia (i.e., loss of one-quarter of the visual field).

The correct answer is: Lower bank of calcarine fissure

Question 95

Not answered

Marked out of 1.00

A patient has spent some time in hospital, and the team asked you to review him as his IQ seems to have deteriorated. The best test to measure the patient's premorbid IQ is

Select one:

- Beck's inventory
- Paired associate learning test
- Rey Osterrieth test
- Luria motor test
- National Adult Reading Test

Your answer is incorrect.

The National Adult Reading Test (NART) is a commonly used clinical method to estimate premorbid intelligence levels in English-speaking patients with dementia and other organic brain lesions. It was developed by Hazel Nelson in 1982. It is a 50-item test which is used for both clinical and research purposes.

The correct answer is: National Adult Reading Test

Question 96

Not answered

Marked out of 1.00

Broca aphasia is also known as

Select one:

- Fluent aphasia
- Non-fluent aphasia
- Conduction aphasia
- Nominal aphasia
- Jargon aphasia

Your answer is incorrect.

Explanation:

Broca aphasia (also called non-fluent or expressive aphasia) had traditionally been characterised by non-fluent speech and impaired repetition but intact auditory comprehension. It has long been thought to be associated with damage to Broca area (i.e. left inferior frontal convolution) or Brodmann area 44.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 48.

The correct answer is: Non-fluent aphasia

Question 97

Not answered

Marked out of 1.00

Which of the following cognitive tests is useful to check semantic memory?

Select one:

- Copying pentagons and colouring them
- Category fluency and digit-symbol test
- Serial subtraction and proverb interpretation
- Object naming and defining the meaning of words
- Letter fluency

Your answer is incorrect.

Semantic memory involves memory for word meaning and general knowledge, the key neural substrate being the anterior temporal lobe. The bedside tests used include object naming and defining the meaning of words.

Patients with semantic breakdown typically complain of loss of words. Vocabulary diminishes, and patients substitute object names with stock-words e.g. ''thing''. There is a parallel impairment in appreciating the meaning of individual words, which first involves infrequent or unusual words.

The correct answer is: Object naming and defining the meaning of words

Question 98

Not answered

Marked out of 1.00

Which one among the following is a neuropsychological theory of ADHD?

Select one:

- Low central coherence
- Reversal learning
- Fronto-striatal executive dysfunction
- Selective attention bias
- Poor external locus of control

Your answer is incorrect.

Explanation:

Attention Deficit Hyperactivity Disorder has been implicated in dysfunction of neural reward circuitry controlling motivation, reward, and impulsivity, including orbitofrontal cortex systems. This condition has increasingly been defined as a disorder resulting from impaired behavioral inhibition leading to executive function deficits. ADHD patients have deficits in higher-level cognitive functions necessary for mature adult goal-directed behaviours, in so-called 'executive functions' (EFs), that are mediated by late developing fronto-striato-parietal and fronto-cerebellar networks. The most consistent findings are in so-called 'cool' EF such as motor response inhibition, working memory, sustained attention, response variability, and cognitive switching as well as in temporal processing, with most consistent deficits in time discrimination and estimation tasks.

Ref: Rubia K. Cognitive neuroscience of attention deficit hyperactivity disorder (ADHD) and its clinical translation. Frontiers in human neuroscience. 2018 Mar 29;12:100.

The correct answer is: Fronto-striatal executive dysfunction

Question 99

Not answered

Marked out of 1.00

Lesion of which lobe causes nominal aphasia?

Select one:

- Frontal
- Cerebellum
- Parietal
- Occipital
- Temporal

Your answer is incorrect.

In nominal aphasia, the patient cannot name objects, although they have plenty of words at their disposal.

Although many experts have believed that damage to either Broca's or Wernicke's area is the main cause of anomia, current studies have shown that damage in the left parietal lobe is the epicenter of anomic aphasia. (Ref: Fridriksson J, Kjartansson O, Morgan PS, et al. (2010). Impaired speech repetition and left parietal lobe damage. J. Neurosci. 30 (33): 11057-61.)

The correct answer is: Parietal

Question 100

Not answered

Marked out of 1.00

Jane experiences sensory aura with tingling, numbness with vague pain sensations, body image distortions, vertigo and visual illusions. The most likely seat of epileptic focus is

Select one:

- Brain stem
- Frontal lobe
- Temporal lobe
- Parietal lobe
- Occipital lobe

Your answer is incorrect.

Explanation:

Seizures arising from the parietal lobe may be difficult to diagnose because of their subjective nature. Positive and/or negative sensory features are common. Sensorimotor phenomenon and vestibular hallucinations suggest onset in the parietal lobe. Paraesthesias, visual hallucinations, visual illusions, somatic illusions, vertiginous features can occur. Seizures arising from the dominant hemisphere can cause receptive language impairment. Parietal lobe complex partial seizures can have auras like epigastric sensations, visual hallucinations, panic attacks, and behavioural arrest. There is an involvement of the other lobes as the seizure spreads. When focal seizures from the parietal lobe spread and involve the temporal lobe, loss of consciousness and automatisms may occur.

The correct answer is: Parietal lobe

Question 101

Not answered

Marked out of 1.00

Which one among the following is one of the performance subscales of WAIS?

Select one:

- Comprehension
- Arithmetic
- Picture completion
- Similarities
- Digit span

Your answer is incorrect.

6 Verbal subtests-information, comprehension, arithmetic, similarities, digit span, vocabulary and letter-number sequencing 5 Performance subtests of the WAIS include digit symbol, picture completion, block design, picture arrangement, object assembly.

The correct answer is: Picture completion

Question 102

Not answered

Marked out of 1.00

A lesion in which of the following sites is known to result in bitemporal hemianopia?

Select one:

- Frontal lobe tumour
- Pituitary tumour
- Temporal lobe tumour
- Parietal lobe tumour
- Occipital lobe tumour

Your answer is incorrect.

Explanation:

Bitemporal hemianopsia (or bitemporal hemianopia) describes the ocular defect that leads to impaired peripheral vision in the outer temporal halves on the visual field of each eye. This condition commonly results from a tumour (e.g. pituitary adenoma) or lesion impinging on the optic chiasm, the decussation point of the optic nerve conveying visual information from the nasal retina in each eye. While most children with pituitary tumours present with the clinical picture of raised intracranial pressure, the majority o f the adults present with visual failure and optic atrophy. Bitemporal hemianopia is fairly frequently found (one-third of patients at the time of diagnosis) but is usually asymmetrical and unpredictable in its evolution.

The correct answer is: Pituitary tumour

Question 103

Not answered

Marked out of 1.00

Alexia without agraphia (pure alexia) is seen in lesions of

Select one:

- Middle meningeal artery
- Posterior cerebral territory
- Anterior cerebral territory
- Middles cerebral territory
- Posterior inferior cerebellar artery

Your answer is incorrect.

Explanation:

Alexia without agraphia (or pure alexia) is characterised by loss of reading abilities in a literate person. They can name things well but cannot read properly. Hence, in pure alexia, the patient has preserved visual recognition and writing skills. Most cases of alexia without agraphia are caused by left posterior cerebral artery occlusion and a resultant infarct of the left visual cortex as well as the splenium of the corpus callosum.

Ref: Rupareliya C et al. Alexia without agraphia: a rare entity. Cureus. 2017 Jun;9(6).

The correct answer is: Posterior cerebral territory

Question 104

Not answered

Marked out of 1.00

A major use of Glasgow Coma Scale is to

Select one:

- Predict mortality
- Predict development of late complications
- Predict time course for recovery
- Predict development of co-morbidities
- Predict development of early complications

Your answer is incorrect.

The Glasgow Coma Scale (GCS) is routinely used in the acute care setting after traumatic brain injury (TBI) to guide decisions in triage, based on its ability to predict mortality.

Although the GCS has been previously demonstrated to predict mortality, efficacy in the prediction of functional outcome is not well established.

GCS less than 8 can mean a mortality rate as high as 70%. In the lower ranges of GCS score, there is a direct linear relation to a poor outcome including severe neurological disability, vegetative state and death.

Balestreri, M., Czosnyka, M., Chatfield, D., Steiner, L., Schmidt, E., Smielewski, P., et al. (2004). Predictive value of Glasgow Coma Scale after brain trauma: Change in trend over the past ten years. Journal of Neurology, Neurosurgery, and Psychiatry, 75(1), 161?162.

The correct answer is: Predict mortality

Question 105

Not answered

Marked out of 1.00

A 43-year-old man can speak normally and comprehend what is spoken. He can also write spontaneously and to dictation, but reading comprehension is impaired.

Select one:

- Pure word deafness
- Jargon aphasia
- Broca's aphasia
- Pure word blindness
- Wernicke's aphasia

Your answer is incorrect.

Pure word blindness (alexia but no agraphia or agnosic alexia) is one form of alexia where patients suffer from severe reading problems while other language-related skills such as naming, oral repetition, auditory comprehension or writing are typically intact.

The correct answer is: Pure word blindness

Question 106

Not answered

Marked out of 1.00

Immediate repetition of 3 items in MMSE is used to test

Select one:

- Recognition
- Recall
- Digit span
- Fluency
- Registration

Your answer is incorrect.

Immediate repetition of 3 items in MMSE is used to test 'Registration'.

Delayed recall can be tested by asking to repeat the items after a distraction test

The correct answer is: Registration

Question 107

Not answered

Marked out of 1.00

Which of the following components of transcribed RNA is removed before translation occurs?

Select one:

- Introns
- Ribosomal attachment sites
- Cap of RNA
- Aminoacid coding portions
- Tail of RNA

Your answer is incorrect.

An intron is any nucleotide sequence within a gene that is removed before generating the final mature RNA product of a gene. Introns are removed by a process called splicing before protein translation begins. The ribonucleoprotein assembly that removes the intervening sequences from pre-mRNAs through splicing is called the spliceosome.

The correct answer is: Introns

Question 108

Not answered

Marked out of 1.00

Which of the following genetic element is implicated in Huntington's disease?

Select one:

- Presenilin-1 gene on chromosome 14
- IT-15 on chromosome 4
- IT-4 on chromosome 15
- APO E4 on chromosome 19
- IT-1 gene on chromosome 1

Your answer is incorrect.

The mutation in the IT-15 gene ("interesting transcript 15") on chromosome 4 with an unstable region of the harbouring an expanded trinucleotide repeat motif, is associated with Huntington's disease.

The correct answer is: IT-15 on chromosome 4

Question 109

Not answered

Marked out of 1.00

The term used for the purpose of identifying the locus of a gene on the chromosome by studying the extent to which it co-segregates with a marker gene is called

Select one:

- Mutagenesis
- Physical mapping
- Association studies
- Linkage analysis
- Polymerase chain reaction

Your answer is incorrect.

Explanation:

Linkage analysis is used to obtain a coarse chromosomal location of the genes associated with a phenotype under investigation (e.g. a genetic disease or an important quantitative trait). All genetic mapping methods aim to identify disease-associated variants based on their chromosomal position and the principle of genetic linkage. Genes that are physically close to one another on a chromosome are genetically linked, and those that are further apart or are on different chromosomes are genetically unlinked. Unlinked genes will recombine at random. Genetic loci that are linked will recombine less frequently than expected by random segregation, with the degree of recombination proportional to the physical distance between them.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 988.

The correct answer is: Linkage analysis

Question 110

Not answered

Marked out of 1.00

Two parents are both affected with the same autosomal recessive disorder give birth to a clinically normal child. Which of the following can explain the above?

Select one:

- Incomplete penetrance
- Variable expressivity
- Pleiotropy
- Locus heterogeneity
- Allelic heterogeneity

Your answer is incorrect.

Locus heterogeneity refers to mutations at several different loci producing identical phenotypes.

A good example is presenilin1, 2 and Beta APP mutations all causing phenotypically similar presenile Alzheimer's dementia.

In this example, each parent may have had a different mutation, so the child is a heterozygote carrier at both loci.

The correct answer is: Locus heterogeneity

Question 111

Not answered

Marked out of 1.00

The term used to describe a point mutation that leads to failure of expression of alleles is

Select one:

- Dominant negative mutation
- Lethal mutation
- Apoptotic function
- Loss of function mutation
- Lesion mutation

Your answer is incorrect.

Explanation:

Loss-of-function mutations are the result of gene product having less or no function. Phenotypes associated with such mutations are most often recessive. Loss-of-function mutations abolish a gene's biomolecular activity. The existence of a category of alleles distinguished by a derived loss of biochemical functions has been described by various names: amorphic, loss-of-function, non-functional, knockout, null, pseudogene, or gene loss. Total gene loss is the most obvious case of loss of function. Yet total gene loss is not the only means by which loss of function can occur. Single mutations and many mutation types such as premature stop codons, frameshifts, splice site disruptions, and elimination of regulatory regions required for gene expression can have effects that are functionally indistinguishable from complete gene loss.

Ref: Monroe JG, McKay JK, Weigel D, Flood PJ. The population genomics of adaptive loss of function. Heredity. 2021 Mar;126(3):383-95.

The correct answer is: Loss of function mutation

Question 112

Not answered

Marked out of 1.00

Following a Robertsonian translocation in phenotypically normal individual which of the following types of chromosomes is produced?

Select one:

- Holocentric
- acrocentric
- Metacentric
- telocentric
- submetacentric

Your answer is incorrect.

Following a Robertsonian translocation, the small 'p' arms are discarded, and a metacentric fusion chromosome results.

The correct answer is: Metacentric

Question 113

Not answered

Marked out of 1.00

A family with three affected children has a higher recurrence risk for a disease than another family with only two affected children. This disease is most likely to be inherited via

Select one:

- Mitochondrial
- Autosomal dominant
- Autosomal recessive
- Multifactorial
- X linked dominant

Your answer is incorrect.

An increasing recurrence risk according to the number of relatives affected is seen in multifactorial diseases. Recurrence risks for Mendelian disorders are unaffected by number of offsprings already affected.

The correct answer is: Multifactorial

Question 114

Not answered

Marked out of 1.00

Which of the following disorders is associated with HLA-DQB1*-0602?

Select one:

- Huntington's disease
- Narcolepsy
- Pick's disease
- Schizophrenia
- Bipolar disorder

Your answer is incorrect.

Explanation:

Narcolepsy is a chronic disorder that is primarily characterised by excessive daytime sleepiness and, in most individuals, cataplexy. The latest edition of the International Classification of Sleep Disorders has reclassified narcolepsy into narcolepsy type 1 (NT1) and NT2. This is based on the finding that the absence of the hypocretin peptides (also known as orexins; which are neuropeptides with a role in the regulation of sleep and wakefulness) is a fundamental marker in most patients with narcolepsy with cataplexy. Patients with NT1 have cataplexy and/or low levels of hypocretin 1 (also known as orexin A) in the CSF, whereas patients with NT2 do not have cataplexy and have normal levels of hypocretin 1. The main genetic risk for narcolepsy is the HLA-DQB1*06:02 allele, which explains the majority of the known genetic effects in NT1. Depending on the population, 86-98% of patients with NT1 carry the HLA-DQB1*06:02 allele.

Ref: Kornum BR et al. Narcolepsy. Nature reviews Disease primers. 2017 Feb 9;3(1):1-9.

The correct answer is: Narcolepsy

Question 115

Not answered

Marked out of 1.00

A patient with schizophrenia is less likely to get married and produce children due to the nature of the illness. This will serve to reduce the gene pool of schizophrenia causing genes. This is called as

Select one:

- Pleiotropy
- Natural selection
- Assortative mating
- Genetic drift
- Negative eugenics

Your answer is incorrect.

Natural selection serves to maintain a low pool of certain disorders. Due to impaired fertility and affliction at reproductive age, schizophrenia genes may be maintained at a lower level of constancy in the population.

The correct answer is: Natural selection

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